# Row-Independent Feature Pre-Screening with Language Models for High-Dimensional Credit Scoring: A Frozen Evaluation on Temporal and Recency-Shift Holdouts

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**Abstract**

Supervised feature selection over engineered credit-scoring universes of hundreds to thousands of predictors is recomputed on up to millions of decisions at every refit. We ask whether a large language model (LLM) reading only the variable book (names, descriptions, lineage; no data rows), at a cost that does not grow with the number of decisions, can pre-screen such a universe before statistical selection. Thirteen strategies are evaluated on Home Credit, LendingClub and Home Credit Model Stability 2024 (up to 1,959 features and 1,526,659 decisions) with logistic regression and CatBoost at budgets of 20 and 40 features, on holdouts later in calendar time (two portfolios) or shifted in applicant recency (one). Pipelines were frozen before final scoring, but the protocol evolved with knowledge of earlier held-out results. The best LLM-assisted and classical selectors (family leaders) are compared as paired differences in area under the receiver operating characteristic curve (AUC), with bootstrap intervals and no hypothesis test, under three leader-choice rules. The LLM-assisted leader is higher in five of six cases under all three rules, by +0.012 to +0.051 (held-out rule) and +0.013 to +0.052 (development rule), and loses less AUC from development to holdout. Home Credit logistic regression favours the classical leader (−0.010 to −0.027). In a matched measurement on one Stability 2024 development fold, mRMR took 522 seconds over the full universe and 21 seconds over a 100-feature pool. Semantic pre-screening is a usable, dataset-dependent first-pass filter that needs stability and drift monitoring.

*Keywords:* Credit scoring, Big data, Feature selection, Large language models, Temporal validation, High-dimensional data, Selection stability

## 1. Introduction

Retail credit-risk models increasingly draw on large engineered feature universes. Application records joined with bureau histories, previous-loan behaviour and transaction aggregates at several depths and time windows yield hundreds to thousands of candidate predictors observed on hundreds of thousands to millions of decisions, many of the predictors redundant, some unstable over time and a few unavailable at the moment a lending decision is taken [1, 2]. The largest dataset in this study holds 1,959 engineered features on 1,157,512 development decisions. Selecting a compact subset is not a purely statistical exercise, and in a regulated domain the subset must also remain reproducible, monitorable and explainable throughout the life of the model [3, 4, 5].

Classical feature selection addresses these requirements from the data side: filters rank variables by relevance and redundancy statistics [6, 7], wrappers search subsets with a learning algorithm in the loop [8, 9, 10], and embedded methods take importance from the fitted model [11, 12]. None of them reads the meaning of a feature name or its data-dictionary description. Large language models (LLMs) have recently been proposed as a complementary source of evidence: given names, descriptions and a task statement, an LLM can rank features without seeing a single training row, and such semantic priors are competitive with data-driven selectors on generic benchmarks [13, 14, 15, 16]. This makes the LLM a candidate *pre-screener* for credit scoring: it reads the variable book and returns a ranked candidate pool that can be used directly or handed to a statistical selector that then operates on a smaller pool.

Scale sharpens the case for such a pre-screen. Supervised selectors are refitted in every validation fold and at every model refresh, and their cost grows with the number of decisions and, for redundancy-aware selectors, with the number of candidates over which redundancy is computed: on the first development fold of Stability 2024, one mRMR pass took 522 seconds over the 1,959-feature universe and 21 seconds over a 100-feature pool. A language model that

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reads the variable book rather than the data rows has a cost that depends on the size of the variable book and not on the number of decisions, and a cached ranking can be reused across refits. This paper asks whether such a row-independent pre-screen can shrink the space that supervised selection must search without giving up held-out discrimination.

Whether these tools help credit scoring with large engineered universes is an empirical question. RQ1: does an LLM pre-screened subset discriminate at least as well as a classically selected one on a held-out population scored once by a frozen pipeline, later in calendar time or shifted in prior-application recency, under a fixed budget? RQ2: are the selected subsets reproducible across development folds, which matters for governance independently of accuracy [17, 18]? One further question, whether aggregating several classical rankings over a wider candidate pool raises held-out discrimination over a single classical filter, is examined in a pre-declared supplementary experiment (Appendix B) and summarised in Section 7.2; Appendix C reports the development evidence for the feature budgets. Existing LLM-selection studies address RQ1 on general benchmarks with random splits and small tables; the two questions have not been examined together on credit portfolios of this size with temporal or recency-shift held-out validation.

This paper reports such an evaluation. Thirteen selection strategies and a full-feature reference are compared on three credit datasets, Home Credit, LendingClub and Home Credit Model Stability 2024 [19, 20, 21]. Each dataset has a held-out population, labelled HO throughout, that is later in calendar time for LendingClub and Stability 2024 and a recency-based covariate-shift holdout for Home Credit (Section 5.1); the frozen final pipelines score it once, but earlier protocol versions inspected it. Two backbones are used, a penalised logistic regression and a CatBoost model [22], fitted on subsets of $K = 20$ and $K = 40$ original features. Every ranking, subset, preprocessing map and hyperparameter is fitted on development data and frozen before the final HO scoring run of each pipeline. The LLM is GPT-4.1 mini [23] under one definition-only information contract shared by the three datasets.

The paper makes four contributions. First, a held-out comparison, temporal for LendingClub and Stability 2024 and recency-shifted for Home Credit, of classical, model-based and LLM pre-screened selection, whose semantic stage reads no data rows, across three credit portfolios of up to 1,959 engineered features and 1.5 million decisions (1,526,659) and two backbones; the six paired HO comparisons between the methods with the highest observed HO AUC in each family are reported as effect sizes with paired bootstrap intervals, and repeated under two further leader-choice rules, one of which uses no held-out information to choose the compared methods. Second, an explicit specification of the information each selection stage may consume, separating target-free from target-aware stages, with the freeze and leakage controls that separate fitting from HO scoring within each protocol version. Third, a separation of discrimination, fold-to-fold reproducibility and distribution drift as distinct outcomes, with a pre-declared classical rank-voting family reported as supplementary material. Fourth, a transparent account of how the evaluation protocol evolved across its versions and which changes coincided with changes in the results. Section 2 reviews related work, Section 3 gives background, Sections 4 and 5 specify the methods and the experimental setup, and Section 6 reports the results in the order of the questions a reader is likely to ask: the headline comparison (Section 6.1), its robustness (Section 6.2), what drives it (Section 6.3), stability, drift and secondary metrics (Section 6.4) and scalability and cost (Section 6.5); Section 7 discusses them and Section 8 concludes.

## 2. Related Work

*Credit scoring and feature selection.* Baesens et al. [24] and Lessmann et al. [3] established the practice of comparing classifiers on several retail portfolios with complementary metrics and statistical tests, and gradient boosting remains a strong tabular baseline [4]. Feature selection for credit scoring has been studied predictively and economically, for example by optimising the subset against expected profit and acquisition cost [25]. These studies use random or stratified splits and do not measure the reproducibility of the subset, the drift of the selected variables over time or the use of semantic information about them.

*Classical selection and its stability.* In wide data it is the cost of selection, its redundancy handling and its stability, not raw relevance, that dominate the practical difficulty [27]. The filter, wrapper and embedded taxonomy [8, 6] organises classical work: mRMR trades mutual information with the target against redundancy [7], information value is the filter of scorecard practice [2], recursive feature elimination [9] and Boruta [10, 26] are wrappers, and sparsity penalties [11] or SHAP attributions [12] give embedded rankings. Selection stability was framed by Kalousis et al. [17], corrected for chance by Kuncheva [28] and unified by Nogueira et al. [18]; ensemble selection and rank aggregation were proposed as routes to more stable subsets [29, 30]. Ballegeer et al. [5] study the stability of post-hoc explanations in credit scoring, whereas we measure the stability of the subset before modelling. Choosing the reported comparison after inspecting the test results is itself a recognised source of optimism: Cawley and Talbot [31] quantify the selection bias in performance evaluation that follows model selection, Jelizarow et al. [32] illustrate over-optimism in method comparisons, and Berk et al. [33] and Taylor and Tibshirani [34] formalise valid inference after selection; Section 5.6 treats the headline comparisons of this paper in that light.

*Language models for feature selection and in credit risk.* Pre-trained language models encode usable priors over variables [13]. Jeong et al. [14] formalised LLM feature selection from names and task descriptions and found GPT-4 selections competitive with LASSO without access to training rows, Li et al. [15] contrasted text-based and data-driven modes, and Zhang et al. [16] used LLM priors to set LASSO penalties; all are evaluated on general tabular benchmarks with random splits. The LASSO comparator of those studies is reported for the present protocol in Section 6.2 and Appendix F. Beyond selection, LLMs have been used to engineer tabular features from column semantics [37, 38, 39] and to classify tabular rows from serialised text [40]; these studies also use general benchmarks and random splits. In credit risk, language models have mainly been used to extract signal from borrower text [35, 36]. We instead use the model as a pre-screener that reasons about the existing engineered variable book, and borrower-level prediction remains the task of tabular classifiers.

*Temporal validation and drift.* Random cross-validation overstates performance on time-ordered data [41, 42], the population stability index (PSI) is the standard drift statistic of credit risk [43], and the Home Credit Model Stability 2024 competition made temporal stability an explicit objective [21]. No prior study evaluates LLM pre-screening and classical selection jointly on large credit portfolios with frozen held-out scoring and fold-level stability; that joint evaluation is the gap addressed here.

## 3. Background

### *3.1. Credit-Risk Feature Selection*

A retail credit model estimates the probability of a defined adverse event within a horizon from information available at the decision date. The engineered universe combines the application, bureau records, the applicant's history with the lender and behavioural aggregates at several depths, so within-source redundancy is built in. Selection must satisfy four partly conflicting requirements: relevance established on development data only; limited redundancy, which inflates variance in logistic regression and complicates attribution; reproducibility, because a subset that changes at every refit is hard to govern and explain to a validator; and decision-time availability with a distribution stable enough to be monitored with PSI [2, 1, 43]. Scale adds a fifth requirement: on a universe of a thousand or more features observed on a million decisions, relevance statistics, and for redundancy-based selectors pairwise redundancy statistics, are recomputed in every fold, so a pre-screening stage that reduces the candidate pool can lower the cost of the downstream selectors; the size of that saving depends on the selector, and one matched measurement is reported in Section 6.5. Accordingly, this study measures discrimination on held-out populations scored by frozen pipelines [41], reproducibility across ordered development folds and drift for each selected feature, and enforces availability before selection by restricting the universe to approved predictors.

### *3.2. Classical Feature Selection*

Let $X_1, \ldots, X_d$ denote candidate features and $Y \in \{0, 1\}$ the event indicator. *Filters* score features by statistics computed from data alone. The information value (IV) of scorecard practice bins a feature and compares the shares $g_b$ and $e_b$ of non-events and events in bin $b$ through the weight of evidence (WOE) [2],

$$\mathrm{WOE}_b = \ln \frac{g_b}{e_b}, \qquad \mathrm{IV} = \sum_b (g_b - e_b)\,\mathrm{WOE}_b. \tag{1}$$

Mutual information $I(X_j; Y)$ is the general-purpose relevance statistic, and mRMR [7] adds greedily the feature that maximises relevance minus its average redundancy with the already selected set $S$,

$$j^\star = \arg\max_{j \notin S} \Big[ I(X_j; Y) - \frac{1}{|S|} \sum_{k \in S} I(X_j; X_k) \Big], \tag{2}$$

with mutual information estimated by nearest-neighbour estimators [44, 45]: the estimator of Kraskov et al. between continuous variables and that of Ross between a continuous feature and the binary target. Principal component analysis is unsupervised but replaces variables by orthogonal combinations rather than selecting them [46]. *Wrappers* evaluate subsets with a learning algorithm, here recursive feature elimination [9] and Boruta [10, 26], and *embedded* methods derive relevance from the fitted model, here through SHAP attributions [12], with sparsity penalties [11] the classical alternative; their implementations are specified in Section 4.3. Because wrappers and embedded methods use the target, they must be fitted strictly inside development folds.

### *3.3. LLM Pre-Screening and Semantic Selection*

Formally, let $\mathcal{F}$ be the engineered universe of $d$ original features, $\mathcal{D}_{\mathrm{DEV}}$ and $\mathcal{D}_{\mathrm{HO}}$ the development population and the held-out population, labelled HO throughout because it is a later calendar period only for LendingClub and Stability 2024 and a recency-based covariate-shift holdout for Home Credit (Section 5.1), and $K$ a fixed budget. A selector is a map $\sigma$ from development data to a subset $S \subseteq \mathcal{F}$ with $|S| = K$, and it is judged on three quantities: the held-out discrimination $\mathrm{AUC}_{\mathrm{HO}}(f_S)$ of a fixed model class $f$ fitted on $S$ using $\mathcal{D}_{\mathrm{DEV}}$ only; the reproducibility of $\sigma$ across development folds, measured by the Nogueira statistic $\hat{\Phi}$ and mean pairwise Jaccard similarity; and the drift of the selected features between $\mathcal{D}_{\mathrm{DEV}}$ and $\mathcal{D}_{\mathrm{HO}}$, measured by PSI. Classical selectors compute $\sigma$ from feature values and labels. Semantic selection instead uses what a feature *is*: names, descriptions, source lineage, aggregation depth and logical type carry the information an experienced analyst uses when reviewing a variable book, and

this first-pass review, a pre-screen of the universe that consumes no data rows, is the task delegated to the language model [14, 15]. The resulting ranking can be used directly (*pure LLM*), as a screen that reduces $\mathcal{F}$ to a pool refined by a statistical selector, or as a completion rule that fills a statistically stable core (Sections 4.4 and 4.5). The ranking is a cached artefact that is target-free itself, although a pipeline containing it is target-aware whenever screening or refinement uses labels, and, where it was requested separately for each fold (Home Credit and LendingClub), its fold-to-fold agreement is reported with the Nogueira statistic and pairwise Jaccard similarity (Section 6.4.1).

### *3.4. Feature-Selection Stability*

Selection stability is the degree to which a selector returns the same subset when the training sample is perturbed [17]. It is measured here by mean pairwise Jaccard similarity [47] and by the chance-corrected estimator of Nogueira et al. [18], which generalises Kuncheva's correction [28] to a universe of size $d$; both are defined in Section 5.5, and the choice of $d$ matters because the chance correction grows as the universe shrinks, so values computed over different universes are not directly comparable. Stability matters in credit risk because a reproducible subset can be documented once and re-validated cheaply, because large fold-to-fold changes signal that the selector tracks sample noise, and because validators expect persistent score drivers. It is distinct from population stability, the drift of a variable measured by PSI, and from the explanation stability of a fitted model [5].

## 4. Methodology

### *4.1. Overall Research Pipeline*

Figure 1 summarises the pipeline, which is executed in the same order for every dataset.

1. **Assembly and outcome definition.** The engineered universe is assembled (529 predictors for Home Credit, 675 for LendingClub, 1,959 for Stability 2024) and the event definition of Section 5.1 applied; for Stability 2024 a prediction-time availability review of 461 source variables approved 434 raw predictors, and on LendingClub the same kind of review removed every post-origination field and the platform's own pricing and grading fields before the 675 features were built from roughly 93 source columns (Section 5.1).
2. **Split and folds.** Each dataset is split along its ordering index, calendar time for LendingClub and Stability 2024 and prior-application recency for Home Credit, into a development population (DEV) and a held-out population (HO), and DEV is divided into five expanding-window folds along the same index, with one index unit omitted between training and validation partitions (same-date decisions stay together in Stability 2024).
3. **Candidate universe and preprocessing.** Every selector, classical and LLM-assisted, works from the same candidate set: the features of the engineered universe with at most 90% missing values on the training partition of the earliest fold, a screen computed once per dataset, held fixed across the folds and using no outcomes. It retains 373 of the 529 Home Credit features, all 675 LendingClub features, none of which exceeds the threshold, and 1,068 of the 1,959 Stability 2024 features. The information-value interval of $0.01 \leq \text{IV} \leq 0.50$ is used within the IV/WOE ranker and as the screening stage of IV then Boruta; the upper bound, the conventional leakage warning of scorecard practice, removed nothing, because no feature of any of the three universes reaches an information value of 0.50 on the earliest training partition. Preprocessing maps are fitted on the training partition of each fold.
4. **Feature selection.** The thirteen strategies of Sections 4.3–4.5 (Table 1), and the classical voting family of Appendix B, produce a subset of $K = 20$ (logistic regression) or $K = 40$ (CatBoost) original features; target-free rankings use no outcome; the semantic ranking is requested separately for each fold partition on Home Credit and LendingClub, and one ranking served every fold on Stability 2024 (Section 4.4); target-aware selectors are fitted per fold.
5. **Model fitting and validation.** The fixed configurations of Section 5.3 are fitted on the subset; fold validation yields discrimination estimates and the five fold-level subsets used for stability.
6. **Freeze and HO scoring.** Everything is refitted on full DEV; rankings, subsets and settings are recorded in a hash-bound manifest; the frozen pipeline then scores the HO population, and no retuning of that pipeline follows within the protocol version (changes between versions are summarised in Section 4.6 and recorded in Appendix E, Table E.1).

The LLM ranking is target-free: the model receives feature definitions and lineage only, and no outcomes, validation statistics or HO information; domain rules, Random $K$, principal components and the missingness screen that defines the candidate set are also target-free. Information value, mRMR, Boruta, recursive elimination, SHAP ranking, the stable-core construction and model fitting are target-aware and confined to DEV training partitions. Leakage controls thus operate at three levels: HO exclusion from every fitting, selection and ranking step, fold-local fitting of every target-aware component, and hash-bound freezing of every artefact before each final HO scoring run.

### *4.2. Data Preprocessing*

Numeric variables are processed by replacing infinite values with missing values, imputing with the training-partition mean and standardising with training-partition

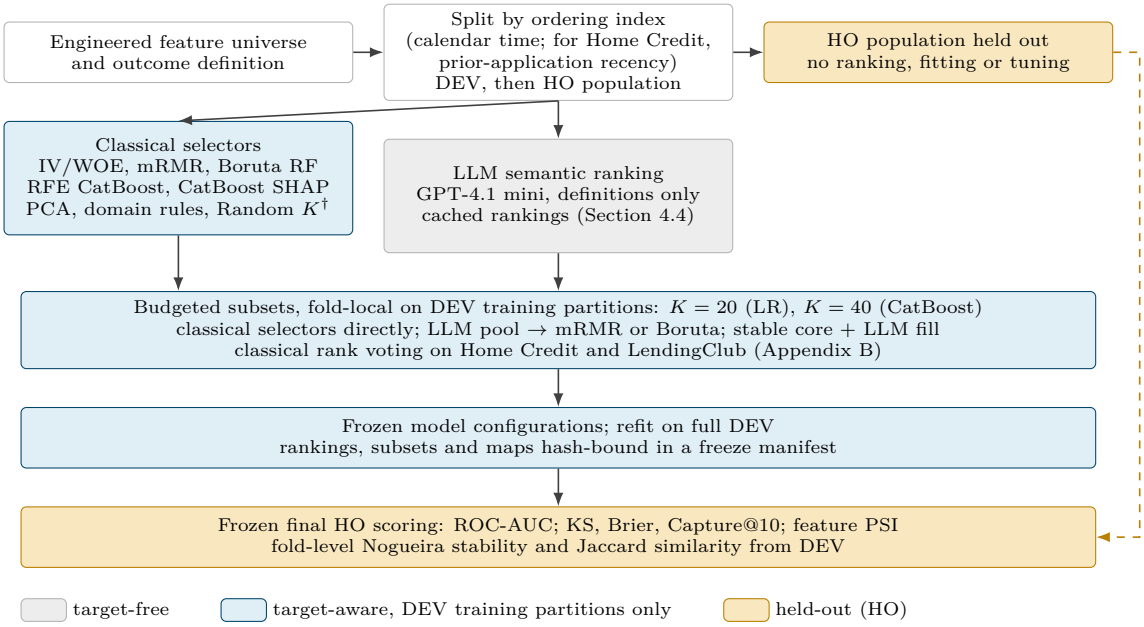



Figure 1: Research pipeline. Grey boxes are target-free stages that consume metadata only; blue boxes are target-aware stages confined to DEV training partitions and refitted once on full DEV, except that PCA, domain rules and Random $K$ (†), grouped with the classical selectors, are themselves target-free (Table 1); orange boxes concern the held-out population (HO), which the frozen final pipelines score once; the HO populations were inspected in earlier protocol versions (Section 4.6), so this is a frozen final scoring run, not a first look. The ordering index of the split is calendar time for LendingClub and Stability 2024 and prior-application recency for Home Credit. Diagram code generated with AI assistance (see the declaration on generative AI).

statistics. Categorical variables receive an explicit missing token and are one-hot encoded; levels unseen in the training partition are ignored, and rare levels are dropped below a minimum frequency of 10 observations for Home Credit and Stability 2024 and 50 for LendingClub. The order of operations is fixed: eligibility screening and selection operate on *original* features, one column per candidate, so a categorical variable is selected or rejected as a whole, and one-hot expansion is performed only after the subset has been selected, so that the budget $K$ counts original variables. Two selectors depart from this rule. Principal-component selection retains components rather than features; and the statistical stage of LLM then mRMR on LendingClub v2 operates on the expanded matrix, so that its budget counts one-hot columns rather than original variables. Appendix F (Table F.3) reports the width of every headline subset, including the one affected here. CatBoost receives this same one-hot expansion rather than its native categorical handling, so that one preprocessing map serves both backbones and the budget $K$ counts original variables under each; Section 6.2 and Appendix F report the three CatBoost cases re-run with native handling. Preprocessing maps are fitted on the training partition of each fold and, for the final model, on full DEV.

### *4.3. Classical Feature-Selection Methods*

Table 1 lists the thirteen strategies and the full-feature reference. All target-aware selectors are fitted on the training partition of each fold with DEV labels only and refitted on full DEV before HO scoring; unless stated otherwise, candidates are ranked by the method's own criterion and the top $K$ original features are kept. *IV/WOE* ranks by Eq. (1) computed on the training partition with ten quantile bins [2], and the same statistic defines the pre-screen of *IV then Boruta*. *mRMR* applies Eq. (2) with mutual information estimated on the training partition [7, 44, 45]; its pool is the whole candidate set when it is used alone and the shortlist when it follows a semantic stage. Throughout the paper *mRMR* denotes this mutual-information implementation; the random-forest/correlation rule used inside the stable-core construction (Section 4.5) and the classical voting family (Appendix B) is called the *RF/correlation filter* and was named legacy mRMR in the early versions of the study. *Boruta RF* confirms features whose random-forest importance (500 trees of depth 6) exceeds the best shuffled shadow importance at $\alpha = 0.05$ with two-step correction and keeps the top $K$ confirmed features [10, 26]. When fewer than $K$ features are confirmed, the subset consists of the confirmed features alone and is flagged as below budget; tentative or rejected features are never added. *RFE CatBoost* removes 20% of the surviving features per step by the importance of an internal CatBoost ranker (500 iterations, depth 6, learning rate 0.05) until $K$ remain [9]. *CatBoost SHAP* keeps the $K$ features with the largest aggregate absolute SHAP contribution over a 10,000-row training sample [12]. *PCA* retains the leading components and is reported as a transformation baseline [46]. *Domain rules* is a deterministic credit-domain scorer that reads no data and calls no model: 42 named-variable rules, family and string-pattern rules for bureau, instal-

ment, credit-card and previous-application aggregates, 15 property-attribute rules and nine semantic-group fallbacks assign each feature a priority, which aggregation suffixes and leakage status adjust, with deterministic tie ordering. The five Home Credit features its leakage rules flag, the aggregates of the count of future instalments in the point-of-sale record, all fail the missingness screen and lie outside the 373-feature candidate set, so no selector could have chosen them. Remaining implementation details are fixed in the public code repository.

Two *diversity controls* are reported in Section 6.2 and Appendix F as tests of the headline comparison rather than as selection strategies. *Family-capped classical* reruns the HO-rule classical leader of each case with a cap of at most $\lceil K/F \rceil$ features per source family, where $F$ is the number of source families in the dataset's universe (6 on Home Credit, 17 on LendingClub and 12 on Stability 2024), the subset filled in the base selector's own ranking order; the caps are therefore 4 and 7, 2 and 3, and 2 and 4 at $K = 20$ and $K = 40$. *Depth-0 classical* restricts the same leader to the depth-0 candidates of Stability 2024, `static_0` and `static_cb_0`. Both are fitted on the training partition of each fold and on full DEV with the frozen backbones of Section 5.3 and scored once on HO, exactly as the selectors are. Where the cap exhausts the base selector's confirmed features before the budget is reached the subset is kept below budget and flagged, as for Boruta. Neither control enters any family, any leader choice or any count in this paper.

### *4.4. Pure LLM Selection*

The language model is GPT-4.1 mini [23], snapshot `gpt-4.1-mini-2025-04-14`, called through the Chat Completions interface with temperature 0 and a JSON response format; it never receives rows, outcomes, validation statistics or HO information. One prompt template, reproduced verbatim in Appendix A, is used for all three datasets under a single *definition-only* information contract. The template carries an evidence-mode placeholder with two admissible values, `definitions_only` and `training_metadata`; every call in this study used `definitions_only`, under which the model may use feature names, definitions, source lineage, aggregation operations and observation windows and no numerical summary of any kind, so missingness, moments, quantiles and cardinalities are withheld as well as every target-conditioned statistic, and the prompt forbids drawing on remembered dataset-specific results. For each candidate the record supplied carries the feature name, its description, source family, lineage formula and semantic group, with the aggregation operation and window of an engineered feature stated in the formula and the name. The descriptions are prepared before ranking from the source-variable definitions, with the aggregation operation and time window of an engineered feature stated in words; their preparation was revised across protocol versions (Appendix E). The descriptions have three provenances. For Home Credit the definition of every source variable is the entry of the data dictionary distributed with the competition data, so no analyst phrasing enters the base definitions; for LendingClub the definitions of the source columns were written by the first author from the platform's column documentation, and every derived feature (ratio, flag, binned version, interaction, missing indicator) is described by a template from the definitions of its source columns; for Stability 2024 the first author wrote the definitions of the approved raw predictors from the provider's documentation. In all three datasets the description of an engineered feature is the base definition with a clause for the aggregation operation and window appended from the lineage by the feature-engineering code, so that, for example, a sum over eleven months is described as the total over eleven months; the template is applied to every feature in the same way and no feature-specific phrasing is added or edited afterwards.

A ranking-budget placeholder fixes the number of names requested, and the prompt asks for exactly that many distinct features ordered from highest to lowest priority, to be refined downstream by the documented budgets. It directs the model to exclude only features whose supplied definition or lineage reveals the outcome, post-decision information or the evaluation split, and to weigh uncertain timing as a concern rather than as leakage. It asks for durable credit meaning, broad semantic coverage and decision-time availability, for avoidance of near-duplicate aggregates of one concept while leaving correlation-based pruning to the statistical selectors, and for alphabetical resolution of remaining ties. When fewer eligible candidates than the budget remain, the model must return status `error` with an empty list rather than a shortened or padded list. Responses are validated strictly: every name must appear verbatim among the candidate records and the array must contain exactly the requested number of distinct names; a failing response is re-requested up to three times and no fallback ranking is used.

The ranking is requested separately for each of the five fold training partitions and for full DEV, and the candidate records supplied are those of the partition. On Home Credit the candidate list is the same 373 of the 529 features in every call, fixed by the missingness screen on the earliest training fold; on LendingClub it is all 675 features, none of which fails the screen; on Stability 2024 it is the 1,068 eligible features (Table 2). On Home Credit six rankings of 100 names were obtained, one per partition, and the budgets of 20, 40, 60 and 100 are their prefixes; on LendingClub a separate call was made for each budget of 20, 40, 60 and 100 names, giving 24 cached rankings; on Stability 2024 one ranking of 100 names was obtained in a single call, accepted at the first attempt. Each response is validated as above, cached and hash-bound, and the cached response rather than a fresh call is the reproducible artefact: the cached orderings all differ from one another, and calls with identical prompt, model and temperature returned different lists, so a fixed model identifier and zero temperature do not guarantee identical outputs (Section 6.4.1). The

Table 1: Feature-selection strategies evaluated. "Labels" states whether the stage that produces the final subset uses DEV outcomes; "Datasets" states where the strategy is evaluated on held-out data. The full-feature model is a reference and is excluded from selector comparisons. The two diversity controls at the foot are tests of the headline comparison (Section 6.2), not strategies of it, and are excluded from every strategy count, every family and every leader choice in this paper.

| Method | Family | Selection role | Labels | Datasets |
|---|---|---|---|---|
| Full features | Reference | No selection; context only | – | All three |
| Domain rules | Classical | Eligibility and policy filters written for the Home Credit variable book | no | All three[a] |
| Random $K$ | Classical | Baseline: uniformly random subset of size $K$, the floor for every selector | no | All three |
| PCA | Classical | Leading components of the standardised candidates (components, not original features) | no | All three |
| IV/WOE | Classical | Ranking by information value, Eq. (1) | yes | All three |
| mRMR | Classical | Mutual-information relevance against mutual-information redundancy, Eq. (2) | yes | All three |
| Boruta RF | Classical | All-relevant random-forest wrapper with shadow-feature comparison | yes | All three |
| IV then Boruta | Classical | Hybrid: IV eligibility screen followed by Boruta confirmation | yes | All three |
| RFE CatBoost | Classical | Model wrapper: recursive elimination driven by CatBoost importance | yes | All three |
| CatBoost SHAP | Classical | Embedded: ranking by aggregated absolute SHAP contribution | yes | All three |
| Pure LLM | LLM-assisted | Top-$K$ names of the cached semantic ranking of the fold | no | All three |
| LLM then mRMR | LLM-assisted | Hybrid: semantic pool of 60 (LR) or 100 (CatBoost), then mRMR | yes | All three |
| LLM then Boruta | LLM-assisted | Hybrid: semantic pool, then Boruta confirmation | yes | All three |
| Stable core + LLM fill | LLM-assisted | Hybrid: bootstrap-stable RF/correlation core completed from the LLM ranking | yes | All three |
| *Diversity controls (Section 6.2; not strategies of the comparison)* | | | | |
| Family-capped classical | Control | HO-rule classical leader re-run with at most $\lceil K/F \rceil$ features per source family | yes | All three |
| Depth-0 classical | Control | HO-rule classical leader restricted to the depth-0 candidates | yes | Stability 2024 |

[a] The domain-rules scorer was written around the Home Credit variable book (Section 4.3); on LendingClub v2 and Stability 2024 it is applied unchanged and is a portfolio-specific baseline rather than a generic domain baseline (Section 7.4).
The pre-declared classical rank-voting family of Appendix B, a supplementary comparison rather than a strategy of this table, covers Home Credit and LendingClub.
$F$ is the number of source families in the dataset's universe: 6, 17 and 12. The family cap is 4 ($K = 20$) and 7 ($K = 40$) on Home Credit, 2 and 3 on LendingClub, and 2 and 4 on Stability 2024.

*pure LLM* selector retains the top $K$ names of the ranking of its partition and budget. Three call structures were therefore used: per-partition calls with a budget of 100 on Home Credit, per-partition and per-budget calls on LendingClub, and a single call on Stability 2024. The token and cost figures of Section 6.5 cover Home Credit and LendingClub only, because the usage of the Stability 2024 call was not recorded.

The contract governs the evidence the model may use, not which variables are eligible. Eligibility is fixed upstream of every selector, by the universe assembly and the missingness screen of Section 4.1, and the same candidate set is offered to the classical selectors and to the language model. The released universes contain demographic fields that fair-lending rules treat as protected attributes in many jurisdictions: applicant gender and family status on Home Credit (`CODE_GENDER`, `NAME_FAMILY_STATUS`) and the sex field of the person table on Stability 2024; age fields appear as well, and their treatment differs across regimes. All of these are categories recorded by the data providers in the released files, not attributes collected or inferred for this study, and the analysis uses them only as released. No exclusion of these fields was added to the prompt, for two reasons. An exclusion applied in the prompt alone would have narrowed the pool for the LLM-assisted family while the classical selectors kept drawing from the full

candidate set, so the two families would no longer have been compared on the same universe; and any change to the prompt invalidates the cached rankings on which every reported LLM-assisted result rests. Where a lender's policy excludes protected attributes, the exclusion belongs in the eligibility screen, where it binds every selector; that screen was not applied in this study, which does not evaluate fairness (Section 7.6), and Section 6.1 records where these fields appear in the headline subsets.

To test whether the ranking depends on recognising feature names, the six Home Credit calls were repeated in two obfuscated arms with everything except the feature records held fixed: the same snapshot (`gpt-4.1-mini-2025-04-14`), temperature 0, JSON response format, the template of Appendix A, the definition-only evidence mode, the same 373 candidates per partition and the same strict validation. In arm A every name was replaced by an identifier `F001`–`F529` drawn from a seeded permutation (seed 20260914) of the universe, the permutation being drawn per partition so that no identifier carries meaning across calls, and every literal occurrence of the original name was removed from the description text as well; the model saw only identifiers and descriptions, and the identifier-to-name table never entered a prompt. In arm B the names were kept and the descriptions were blank. Each arm yields six rankings of 100 identifiers, decoded and carried through the frozen downstream pipeline exactly as the named rankings are: top 20 to logistic regression and top 40 to CatBoost, five fold models plus the full-DEV refit with the fixed configurations of Section 5.3 (seed 42), scored on the same 120,053 HO rows. The outcome is reported in Section 6.3.1.

To test whether the results depend on how the descriptions are phrased, every description in the three candidate sets was rebuilt from lineage alone by a fixed template stating the raw table, raw variable, depth, aggregation operation, window and type rule of the feature, with no statement of credit meaning; the name, source family, lineage formula and semantic group of each record were unchanged. The ranking calls were repeated under the call structure, template, snapshot, temperature, evidence mode, candidate sets and validation of the cached rankings, and the new rankings were carried through the frozen downstream pipelines to the same HO observations: pure LLM in all six cases, and the LLM-assisted leader of the two cases in which it is not pure LLM, stable core plus LLM fill on Home Credit with logistic regression and LLM then mRMR on LendingClub v2 with CatBoost. The outcome is reported in Section 6.3.2.

### *4.5. Hybrid Selection Methods*

Three hybrids combine the cached semantic ranking with target-aware refinement. *LLM then mRMR* takes the top 60 (logistic regression) or top 100 (CatBoost) features of the ranking as the pool and applies mutual-information mRMR on the fold training partition to reach $K$. The semantic stage is intended to remove variables an analyst would consider implausible or redundant in meaning, and the statistical stage variables redundant in the data; whether the two stages together beat mRMR alone depends on the backbone on Home Credit (Section 6.1). *LLM then Boruta* applies Boruta confirmation to the same pool and truncates to the budget. *Stable core + LLM fill* draws five bootstrap resamples of the fold training partition (80% with replacement, seeds 42 to 46) and runs the RF/correlation filter on each. Features selected in at least four of the five resamples form the stable core, ordered by selection frequency and mean rank; the subset is then completed from the LLM ranking in rank order, without padding from unranked features, so that the core preserves reproducibility while semantically prioritised variables fill the remaining budget. None of these pipelines is target-free as a whole, and the fold-to-fold stability of a hybrid subset depends on both the per-fold semantic ranking and the fitted target-aware stage.

### *4.6. Development of the Protocol*

The protocol above is the final form of an evaluation developed through successive versions; the final results supersede the earlier ones. Each version froze its pipelines before scoring its held-out population, but later versions were designed with knowledge of earlier held-out results, so the final evaluation is a frozen scoring run on held-out populations inspected in earlier versions. Across versions the comparator set changed from a single mRMR baseline to the classical family leader of each case; the feature descriptions supplied to the language model were revised (their preparation is described in Section 4.4); mutual-information mRMR replaced the RF/correlation filter as the stand-alone statistical selector; fold-level rank tests gave way to paired bootstrap intervals without significance claims, the development rule being specified after the HO-rule results were known; and the Stability 2024 cutoff was moved once, from 2020-02-26 to 2020-02-01 (Section 5.2). The full account and Table E.1 are in Appendix E.

## 5. Experimental Setup

### *5.1. Datasets*

Three public credit datasets released through Kaggle are used (Table 2), and Figure 2 shows the scale of the selection problem in each: the decisions on which every supervised selector is refitted, the engineered universe, the screened candidate set, the semantic pool and the budget. *Home Credit* is the Home Credit Default Risk data [19]: the unit is a loan application, the event is the provider's binary target (payment difficulties), and the universe contains 529 original features derived from the application record and the auxiliary tables. Its ordering index is `DAYS_DECISION`, the number of days between the applicant's most recent previous application and the current application, so the split is a recency-based covariate-shift

holdout rather than a calendar-time split of the current applications. DEV holds applicants whose most recent previous application is 241 to 600 days old and HO those with a more recent one. Features that are functions of this index, the previous-application decision days and their aggregates, are not in the candidate set of any selector, so no selector could learn on a range of the index that HO does not contain. Applications without a previous application within the 600-day window, about 29% of the 307,511 applications in the public release, are excluded, which limits the population to which the Home Credit results apply. Every selector, classical and LLM-assisted, selects from the 373 features that pass the 90% missingness screen on the earliest training fold; the budget of 20 or 40 features is the same for all.

*LendingClub v2* is the LendingClub loan book [20]: only resolved 36- and 60-month loans through the 2019-05-01 snapshot are retained, with charged-off or defaulted loans as events and fully paid loans as non-events. The 675 features were generated deterministically from roughly 93 distinct columns of the loan book, and every one of them is an attribute of a single loan and its applicant at origination: 116 raw columns carried through unchanged, 180 binary flags, 158 ratios, 88 missing indicators, 78 further engineered variables, 27 binned versions, 19 interactions and 9 categoricals. No feature aggregates across loans or over calendar periods, and the only elapsed-time quantity is the age of the credit file at issue, the number of months between `issue_d` and `earliest_cr_line`, which is an attribute of the loan rather than a time index. A prediction-time availability review removed two groups of source columns before any feature was built. The first is every post-origination field, among them `total_pymnt`, `recoveries`, `last_pymnt_amnt`, `out_prncp`, the `last_fico_range` fields, `hardship_flag` and `debt_settlement_flag`, which record what happened after the credit decision and would reveal the outcome. The second is LendingClub's own pricing and grading fields, `int_rate`, `grade`, `sub_grade`, `installment` and `funded_amnt`, which encode the platform's own assessment of the application: excluding them poses the selection problem over applicant and bureau attributes rather than over a re-encoding of a credit decision already taken, at the cost that the AUC levels reported here are not comparable with LendingClub studies that retain the platform's grade or interest rate. Several credit-file attributes of the loan book were introduced by the platform during the sample period, so their missingness is higher in DEV than in HO, and the features derived from them inherit that pattern; the highest missingness rate reached by any LendingClub feature on the earliest training partition is 82%, so the screen that defines the candidate set retains all 675; 566 features fall below 40%, two lie between 40 and 60% and 107 between 60 and 82%. None of them appears anywhere in the cached semantic rankings, so no LLM-assisted subset contains one; the target-aware selectors compute their criteria on the DEV training partition, where a field that is sparse in that period carries little information value, mutual information or model importance and is down-weighted by construction. Both the language model and the classical selectors select from the same 675 features. The ordering index is the number of days from the issue month to the last finalised issue month, 2018-12-01, so the HO window is later in calendar time. DEV comprises the loans issued from January 2014 to December 2015 and HO the loans issued in 2016. Thirty-six-month loans account for 445,596 of the 598,649 DEV loans and 232,361 of the 293,105 HO loans, with event rates of 14.5% (36-month) and 34.3% (60-month) in DEV and 19.8% and 36.4% in HO, so the DEV–HO difference in event rate arises within both terms and not only from a change in term mix.

*Stability 2024* is the Home Credit Credit Risk Model Stability data [21]: the unit is a credit decision dated by `date_decision` and the event is the provider's default target. The 1,959 engineered features were generated deterministically from the 434 approved raw predictors. The training base table defines the population and the case identifier. At depth 0 the two static families, `static_0` and `static_cb_0`, are joined one-to-one to the base. At depth 1 ten table families with several rows per case, `applprev_1`, `tax_registry_a_1`, `tax_registry_b_1`, `tax_registry_c_1`, `credit_bureau_a_1`, `credit_bureau_b_1`, `other_1`, `person_1`, `deposit_1` and `debitcard_1`, are reduced to one row per case with fixed outcome-independent operations: count, mean, minimum, maximum, standard deviation and the chronologically latest value for numeric and date variables, and mode, most recent category and number of distinct values for categorical variables. Date variables are expressed relative to `date_decision` where appropriate and future information is excluded; depth-2 tables are excluded; no ratios, interactions or target-informed transformations are introduced; and every generated variable keeps its lineage to the raw table, raw variable, depth, operation and type rule. Time windows, where they appear, come from the names of the raw predictors rather than from the engineering. Of the 1,959 features, the 1,068 with at most 90% missing values on the earliest training fold form the LLM candidate set. Home Credit and Stability 2024 share institutional lineage, whereas LendingClub is a distinct peer-to-peer platform, which matters for the transfer experiments.

### *5.2. Held-Out DEV/HO Design*

Each dataset is partitioned along its ordering index into a development population (DEV) and a held-out population (HO). For LendingClub and Stability 2024 the index is calendar time and HO is later than DEV, which addresses whether a selected subset and a fitted model still rank risk correctly on loans issued or decisions taken after development; random splits would mask both population drift and the optimism of subsets tuned on the same period [41, 42]. The split orders decisions, not outcomes: the provider-defined outcome horizons are not fully specified (Section 7.6), so the design is a chronological holdout and

Table 2: Datasets, held-out partitions and feature universes. The candidate set is the screened universe from which every selector chooses, fixed on the earliest training fold. HO labels the held-out population of each dataset. Home Credit windows are days of the most recent previous-application decision relative to the current application, a recency-based covariate-shift holdout rather than a calendar-time split; LendingClub windows are days from the issue month to 2018-12-01; Stability 2024 windows are calendar dates of `date_decision`.

| | Home Credit | LendingClub v2 | Stability 2024 |
|---|---|---|---|
| Observation unit | Loan application | Issued loan | Credit decision |
| Event definition | Payment difficulties (provider target) | Charged off or default; resolved loans only | Default (provider target) |
| DEV window | day −600 to −241 | day −1795 to −1096 (issued 2014-01 to 2015-12) | 2019-01-01 to 2020-01-31 |
| DEV $N$ | 99,092 | 598,649 | 1,157,512 |
| DEV event rate | 7.93% | 19.54% | 3.16% |
| HO window | day −240 to −1 | day −1065 to −730 (issued 2016-01 to 2016-12) | 2020-02-01 to 2020-10-05 |
| HO $N$ | 120,053 | 293,105 | 369,147 |
| HO event rate | 8.90% | 23.29% | 3.09% |
| Feature universe | 529 original features | 675 engineered features (93 source columns) | 1,959 engineered features |
| Candidate set, all selectors | 373 of 529 (≤90% missing) | 675 of 675 (≤90% missing) | 1,068 of 1,959 (≤90% missing) |

The Stability 2024 column describes the cohort of the headline evaluation (cutoff 2020-02-01); the earlier cohort used during protocol development (cutoff 2020-02-26) is described in Section 5.2.

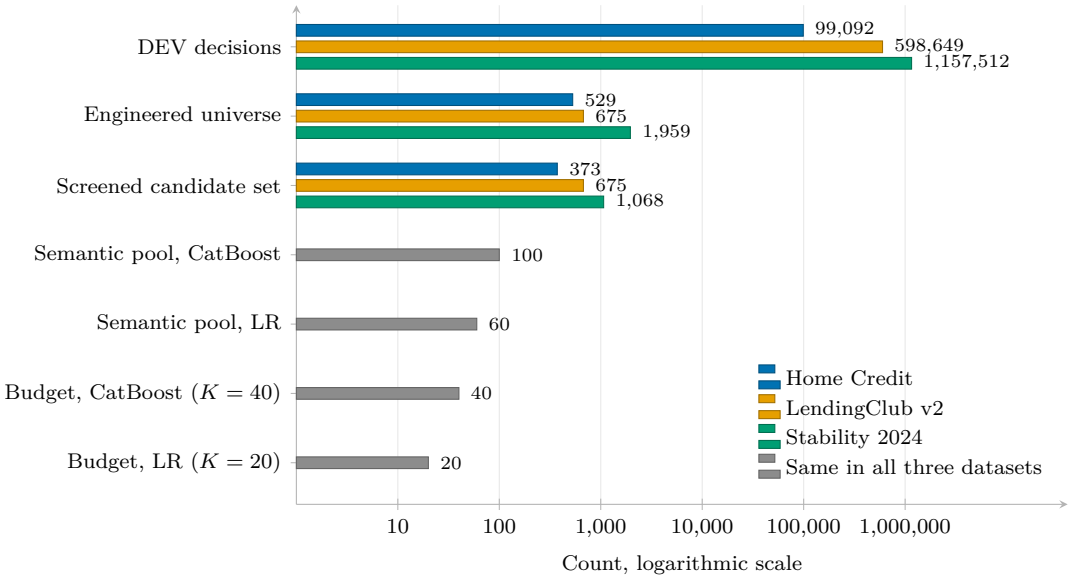



Figure 2: Scale of the selection problem. For each dataset: the development (DEV) decisions on which every supervised selector is refitted; the engineered universe; the screened candidate set offered to every selector, classical and LLM-assisted (features with at most 90% missing values on the earliest training partition); the semantic pool that the hybrid pipelines hand to their statistical stage (60 names for logistic regression, 100 for CatBoost); and the budget of the final subset ($K = 20$ and $K = 40$). The pool and the budget are the same in all three datasets. Logarithmic axis. Plot code generated with AI assistance (see the declaration on generative AI).

does not by itself establish that every development label would have been observable at the held-out decision dates, as a fully simulated deployment would require. For Home Credit the index is prior-application recency, so HO is a recency-based covariate-shift holdout of applicants with a more recent previous application, not a set of later applications. The event rate differs between the two windows, 7.93% in DEV against 8.90% in HO (Table 2), so the Home Credit holdout asks the frozen pipelines to rank risk in a subgroup that is both differently distributed and riskier than the development population, a harder target than a shift in covariates alone. The split point was fixed on the index rather than on a target share of the population, so the Home Credit HO population (120,053 applications drawn from a 240-day recency window) is larger than DEV (99,092 from a 360-day window): applicants whose most recent previous application is recent are denser in the release than those whose most recent one is old. The or-

dering constraint was preferred to an equal split because moving the cut to balance the two sides would have set it with knowledge of the population it defines. Within DEV, five expanding-window folds are formed along the same index, each training partition consisting of the lowest-index observations and the validation partition of a later block, with one index unit, one index day for Home Credit and LendingClub and one decision date for Stability 2024, assigned to neither partition. This omitted unit is not a calendar embargo: LendingClub is indexed by issue month, so on that dataset it separates the partitions by position in the ordering and not by elapsed time, and it is the ordered split itself that keeps later loans out of training. For Home Credit these folds are ordered by recency rather than by application date, so the expanding window enlarges the training partition along the recency index for consistency with the other two datasets rather than to simulate the passage of time. Every component is fitted on the training partition, applied to the validation partition without refitting, and refitted once on full DEV after development.

For Stability 2024 the HO cutoff of the headline evaluation is 2020-02-01: DEV comprises the 1,157,512 decisions dated 2019-01-01 to 2020-01-31 (event rate 3.16%) and HO the 369,147 decisions dated 2020-02-01 to 2020-10-05 (3.09%), 24.2% of the 1,526,659 decisions. During protocol development the cutoff was 2020-02-26 (DEV 1,221,743 decisions at 3.24%; HO 304,916 decisions with 8,349 events, 2.74%). It was moved 25 days earlier for the final evaluation, after the earlier run had been scored, because the later tail compared the development population with a materially lower-risk HO population. The LendingClub HO window runs from day $-1065$ to day $-730$ (loans issued from 2016-01 through 2016-12), and only loans resolved by the 2019-05-01 snapshot are evaluated; loans unresolved at the snapshot are excluded. This restriction can preferentially retain early defaults and early repayments when other loans of the same cohort remain unresolved, and follow-up duration differs across issue cohorts and between 36- and 60-month terms, so population composition and the reported DEV–HO event-rate difference (Table 2) should be read with this selection in mind, and the LendingClub findings describe this resolved-loan population. Before each final scoring run, the feature universe, the cached LLM rankings, every selected subset, the preprocessing maps, and the model configurations and fitted models are frozen, and HO rows enter only the computation of the reported metrics. The Stability 2024 event rates informed the cutoff described above (Section 4.6).

### 5.3. Predictive Models

Two backbones are used in every dataset–selector case, spanning a linear scorecard-style model and a non-linear boosted model [3, 4], with configurations fixed and identical across selectors. The logistic regression uses the `liblinear` solver of scikit-learn [48] with an L2 penalty and $C = 1$, at most 1,000 iterations, balanced class weights and random seed 42. CatBoost [22] uses depth 10, learning rate 0.01, L2 leaf regularisation 95, minimum 290 observations per leaf, a feature-sampling rate (`rsm`) of 0.90, random strength 0.125, depthwise growth, Bernoulli bootstrap with subsample 0.55, the Logloss objective with AUC evaluation, balanced class weights, seed 42, four threads and a fixed 1,500 iterations without early stopping, so that no validation information enters the fitted model. Balanced class weights are used because event rates range from about 3% to 23%. The same two configurations are applied to every feature count, from the 20- and 40-feature subsets to the full-feature reference models on 373, 675 and 1,959 features, the Stability 2024 references being fitted on the unscreened 1,959-feature universe; the CatBoost configuration was carried over unchanged from the authors' prior credit-scoring work at the Central Bank of the Republic of Uzbekistan on a separate, non-public portfolio, where it had been tuned on a validation set, and the logistic-regression configuration is a standard L2-penalised setting with $C = 1$. No hyperparameter of either backbone was tuned on any partition of the three datasets studied here, and one configuration per backbone serves all three. The full-feature reference was not tuned separately, so the "Full" column of Table 3 is a context model under the compact configuration, not a tuned upper bound for the full universe.

### 5.4. Feature Budgets

The budgets are $K = 20$ original features for logistic regression and $K = 40$ for CatBoost. They were determined in DEV-only experimentation and frozen before HO evaluation. Larger linear subsets improved development discrimination only marginally, on average $+0.004$ development AUC at $K = 30$ and $+0.005$ at $K = 40$ for one and a half to two times the feature count, so $K = 20$ was fixed as the compact operating point; CatBoost continued to benefit up to $K = 40$, gained nothing further beyond it on average, and was less sensitive to redundant predictors. Appendix C reports the development curves and shows that the budget sets the level of every selector in a case without changing the ordering between the two families, except at $K = 10$ on Stability 2024 with logistic regression. The budgets are model-specific rather than dataset-specific, so selectors within a backbone are compared at equal subset size, whereas comparisons between backbones differ in both model class and budget. Because $K$ counts original variables (Section 4.2), equal subset size does not mean equal input width once categoricals are expanded; Appendix F (Table F.3) gives the column count each headline subset delivers to its model.

### 5.5. Evaluation Metrics

Let $y_i \in \{0, 1\}$ be the event label of HO observation $i$, $p_i$ its predicted event probability, $N_+$ and $N_-$ the numbers of events and non-events, and $N = N_+ + N_-$.

*ROC-AUC.* The area under the receiver operating characteristic curve is the probability that a randomly chosen event scores higher than a randomly chosen non-event, with half credit for ties [49],

$$\mathrm{AUC} = \frac{1}{N_+ N_-} \sum_{i:\, y_i=1} \sum_{j:\, y_j=0} \big[\mathbf{1}(p_i > p_j) + \tfrac{1}{2}\mathbf{1}(p_i = p_j)\big]. \tag{3}$$

Higher is better. ROC-AUC is threshold-free and is the sole headline metric; the Gini coefficient $2\,\mathrm{AUC} - 1$ adds no information.

*KS.* The Kolmogorov–Smirnov statistic is the maximum vertical separation between the cumulative score distributions of events and non-events,

$$\mathrm{KS} = \max_t \big[\mathrm{TPR}(t) - \mathrm{FPR}(t)\big], \tag{4}$$

which equals the maximum Youden index over thresholds [50]; higher is better. KS is reported as the maximum HO separation. A KS-optimal operating threshold is also recorded in each freeze manifest for deployment, but no threshold-dependent metric is reported, so the threshold plays no part in the results.

*Brier score.* The Brier score is the mean squared error of the predicted probabilities [51],

$$\mathrm{Brier} = \frac{1}{N} \sum_{i=1}^{N} (y_i - p_i)^2, \tag{5}$$

lower is better. It measures the overall squared error of the predicted probabilities, which combines calibration and resolution, rather than ranking; a constant prediction equal to the event rate $\pi$ attains $\pi(1-\pi)$, and log loss, $-N^{-1}\sum_i [y_i \ln p_i + (1-y_i)\ln(1-p_i)]$, is a supporting diagnostic. Because both backbones use balanced class weights and no recalibration (Section 5.3), their outputs are not population event probabilities, so Brier values compare selectors within a dataset and do not indicate deployment-ready calibration.

*Capture@10 and Lift@10.* Let $T$ be the set of observations in the highest-scored decile, a fraction $q = |T|/N \approx 0.10$ of the population. Capture@10 is the share of all events found in that decile and Lift@10 the ratio of the decile event rate to the portfolio event rate,

$$\begin{aligned} \mathrm{Capture@10} &= \frac{\sum_{i\in T} y_i}{\sum_i y_i}, \\ \mathrm{Lift@10} &= \frac{|T|^{-1}\sum_{i\in T} y_i}{N^{-1}\sum_i y_i} = \frac{\mathrm{Capture@10}}{q}; \end{aligned} \tag{6}$$

higher is better, and Lift equals ten times Capture only when the decile is exact.

*PSI.* The population stability index compares a reference distribution with a later distribution over frozen states $b$,

$$\mathrm{PSI} = \sum_b \big(o_b - r_b\big) \ln \frac{o_b}{r_b}, \tag{7}$$

where $r_b$ and $o_b$ are the reference (DEV) and HO proportions in state $b$ [2, 43]; lower indicates less shift. PSI is reported for the selected features only. A score PSI computed in earlier versions compared out-of-fold DEV scores of the five fold models with HO scores of the full-DEV refit, so it mixed scoring-model differences with population change, and it is not reported. For *feature PSI* the states are type-aware: DEV quantile bins plus a missing state for continuous variables, explicit 0/1/missing/unexpected states for binary and one-hot variables, exact DEV levels for discrete variables, and DEV categories plus missing and unseen states for source categoricals. Bins and states are learned on DEV, frozen and applied unchanged to HO; mean, median and maximum feature PSI over the selected subset summarise drift within the chosen inputs. A feature PSI of zero means that no distribution difference was detected under the implemented states, binning and numerical precision, not that the underlying distributions are identical. The conventional references of 0.10 and 0.25 are descriptive monitoring thresholds, not critical values [43].

*Nogueira stability.* Let $Z_{fj} \in \{0,1\}$ indicate whether feature $j$ of a candidate universe of size $d$ is selected in the $f$-th of $B$ selected subsets, $f = 1, \ldots, B$. Throughout the paper the subsets are the five development folds, $B = 5$, except in the repeated-call measurement of Section 6.4.1, where they are ten responses to one fixed prompt, $B = 10$. With $\hat{p}_j = B^{-1}\sum_f Z_{fj}$, $s_j^2 = \frac{B}{B-1}\hat{p}_j(1-\hat{p}_j)$ and $\bar{k} = \sum_j \hat{p}_j$, the Nogueira estimator is [18]

$$\hat{\Phi} = 1 - \frac{d^{-1}\sum_{j=1}^{d} s_j^2}{(\bar{k}/d)\,(1-\bar{k}/d)}. \tag{8}$$

It equals 1 when the same subset is selected in every fold, is chance-corrected so that random selection gives values near 0, and can be negative. The universe $d$ is the pool from which the fold-local selector chose: the screened candidate set (373, 675 or 1,068 features) for the classical selectors, the LLM hybrids and the pure LLM lists. $\hat{\Phi}$ is reported for selectors refitted in each fold and, over the same candidate set, for the pure LLM lists, which were requested separately per fold rather than refitted; because the candidate list supplied is the same in every call (Section 4.4), a fold-to-fold difference between those lists reflects the sampling of the language model at a fixed prompt rather than any change in what it was asked to rank. Section 6.4.1 reports a direct measurement of that sampling on Stability 2024.

*Jaccard similarity.* For fold subsets $S_a$ and $S_b$ the Jaccard similarity is $J(S_a, S_b) = |S_a \cap S_b|/|S_a \cup S_b|$ [47], and mean

pairwise Jaccard averages $J$ over the $B(B-1)/2$ unordered pairs of subsets, ten pairs for the five folds and forty-five for the ten repeated responses of Section 6.4.1; higher indicates more reproducible selection. For the fold subsets the minimum and maximum of $J$ over the ten pairs are reported beside the mean as the fold-pair spread. Unlike Eq. (8) it is not chance-corrected and does not depend on $d$, so the two statistics are reported together.

### 5.6. Statistical Inference

Two sets of six paired comparisons are reported, one for each combination of dataset and backbone. In the first, descriptive set (Table 3), the LLM-assisted selector with the highest observed HO AUC in its family, called the LLM-assisted family leader, is compared with the classical selector with the highest observed HO AUC in its family, the classical family leader. The method matrices, feature budgets and procedure were fixed before HO scoring, but the identities of these two methods were determined from the HO results themselves; no hypothesis test is reported for this set and no significance is claimed for it. In the second, development-rule set (Table 4), each family's leader is chosen on development data only, as the selector with the highest mean validation AUC over the five DEV folds, and the two DEV leaders are compared once on HO. Because the holdout plays no part in choosing what is compared, this set is free of the selection effect of the first; it is nevertheless reported in the same form, as effect sizes with intervals and without a test, for the reason given at the end of this section. In both sets the two pipelines are scored on identical HO observations paired by observation identifier (`SK_ID_CURR` for Home Credit, a stable identifier derived from the processed source row for LendingClub, the decision identifier for Stability 2024), and the paired HO ROC-AUC difference, LLM-assisted minus classical, is reported with a 95% interval from 2,000 target-stratified paired bootstrap resamples that draw the same observation indices for both pipelines [53] with seed 20260721. The interval describes the sampling variability of the difference between the two compared pipelines on the HO population; in the descriptive set it does not account for the selection of those pipelines, and in neither set does it account for the sampling of the language model at a fixed prompt, because the resampling draws HO observations while the cached semantic ranking, and therefore the selected subset of the LLM-assisted pipeline, is held fixed in every resample. That component is measured separately for Stability 2024 in Section 6.4.1 and is reported alongside these intervals rather than folded into them, so that the observation-level and response-level variation remain distinguishable. Comparisons against the full-feature reference and among secondary metrics are point estimates. Because large HO samples make small differences detectable, effect sizes are always reported with the intervals [55], and a paired AUC difference of at least 0.010, two Gini points, is treated as material; smaller differences are reported but are not counted as a result in either direction. The bar applies to a difference computed from one cached semantic ranking; where repeated calls are available the same bar is applied to the spread of the LLM-assisted AUC across responses, a spread below 0.010 being treated as immaterial response sampling.

No $p$-value is attached to either set of headline comparisons. A paired DeLong test, like the paired bootstrap, treats the two fitted pipelines as fixed and asks how the AUC difference would vary if the HO observations were resampled; with 120,053 to 369,147 paired observations that question is answered to the fourth decimal and every headline difference would be significant at any conventional level, so the test would report the size of the holdout rather than anything about the methods. The variation that bears on the claim of this paper lies elsewhere: across portfolios, of which there are three from at most two independent lenders (Section 7.6); across prompts, of which there is one; across responses of the language model, of which there is one per case, ten on Stability 2024 (Section 6.4.1); and across seeds, of which there is one. None of these is resampled by an observation-level test, and the development rule, although it chooses the compared methods without the holdout, was itself specified after the HO-rule results were known (Section 4.6). Both sets are therefore read against the materiality bar and the interval, and the response-level variation is reported beside them where it was measured. The two-sided paired DeLong test [52] with Holm's step-down adjustment [54] is used in one place only, the supplementary classical voting family of Appendix B, whose twelve comparisons were fixed in a protocol file before any held-out result of that family existed.

### 5.7. Leakage and Reproducibility Controls

HO rows are excluded from eligibility screening, feature selection, ranking construction and hyperparameter choice; all target-aware components are fitted fold-locally inside DEV and refitted once on full DEV. The language model receives feature definitions and lineage only, and no statistic computed on any row, validation or HO. Cached rankings, selected subsets, preprocessing maps and model settings are frozen and hash-bound before HO scoring. These controls apply within each protocol version; the development of the protocol across versions used knowledge of earlier held-out results (Section 4.6). Randomness is controlled by fixed seeds (42 for both classifiers and the selectors, 42 to 46 for the stable-core resamples, 20260721 for the inference bootstrap), and the LLM is called with temperature 0 and a fixed snapshot with its rankings cached. Temperature 0 with a fixed snapshot does not make the response deterministic: it is the caching of the rankings, not the decoding temperature, that makes the reported LLM results reproducible. To quantify the residual variation the Stability 2024 full-DEV ranking call was repeated ten times at temperature 0 on the same 1,068 candidate records, and each response was carried through the frozen downstream pipeline (Section 6.4.1). Experiments ran in Python 3.13.5 with scikit-learn 1.8.0, CatBoost 1.2.10 and

Boruta 0.4.3 on a laptop with an Intel Core i7-13620H processor (10 cores) and 40 GiB of memory, on the CPU with four estimator threads, one experiment and one fold at a time. Datasets are public, and code, configuration files and experiment artefacts are available in the repository listed in the Data availability statement.

## 6. Results

### *6.1. Headline comparison*

Table 3 and Figure 3 report the six comparisons that address RQ1. In each dataset–backbone case the LLM-assisted and the classical family leader, the selector with the highest observed HO AUC in its family, are compared on identical HO observations; both leaders were chosen on the HO results, so the comparison is descriptive (Section 5.6); the development-rule comparison follows below. The LLM-assisted leader has the higher held-out ROC-AUC in five of the six cases and the classical leader in the sixth. The LLM-assisted leader is pure LLM in four cases (Home Credit CatBoost, LendingClub v2 LR, Stability 2024 LR and Stability 2024 CatBoost), LLM then mRMR in one (LendingClub v2 CatBoost) and stable core plus LLM fill in the remaining case, Home Credit LR, where the classical selector leads.

The pattern is dataset- and model-specific. LendingClub v2 and Stability 2024 favour LLM-assisted selection under both backbones, by +0.013 to +0.051 for the cached rankings, and on Stability 2024 the margin stays positive in all ten repeated calls under both backbones (Section 6.4.1). Home Credit is mixed: with CatBoost, pure LLM exceeds the RFE CatBoost wrapper by +0.012, whereas with logistic regression mutual-information mRMR exceeds stable core plus LLM fill by 0.021. All five positive differences clear the materiality bar of 0.010 (Section 5.6) in point estimate under each of the three leader rules, the smallest being +0.012. On Stability 2024, where the effect of the semantic response was isolated directly, two of the ten logistic-regression repeats fall below the bar and none of the CatBoost repeats does, while the sign is unchanged in all twenty refits (Section 6.4.1). The Home Credit CatBoost interval under the HO rule, $[+0.009, +0.015]$, is the only one of the five that includes values below the bar. The Home Credit logistic-regression difference is material under the HO rule (0.021); under the development rule it sits at the bar (0.010, interval $[-0.013, -0.008]$). The classical leader changes identity across cases (mRMR, RFE CatBoost, IV then Boruta or CatBoost SHAP), so a single classical baseline would have represented classical selection differently in each case; the classical family contains nine selectors and the LLM-assisted family four. The pattern does not depend on comparing family leaders: excluding the three selectors declared as floors (domain rules, Random $K$ and PCA), the mean HO AUC of the remaining six classical selectors is below the mean of the four LLM-assisted pipelines in the same five cases, by 0.010 to 0.026 (Table 5), and above it on Home Credit logistic regression, by 0.007. An L1-penalised selector run afterwards, CatBoost's native categorical handling, the LendingClub resolution filter and the source concentration of the subsets are examined as robustness checks in Section 6.2, and the input width each headline subset delivers to its model, which $K$ does not fix, is reported in Appendix F (Table F.3).

Choosing each family's leader on the development folds instead (Table 4) changes the compared pair in five of six cases but not the outcome: the LLM-assisted leader is higher in the same five cases, by +0.013 to +0.052, and the classical leader keeps Home Credit logistic regression by 0.010. The rule makes IV then Boruta the classical leader on both Home Credit cases and RFE CatBoost on both LendingClub v2 cases and on Stability 2024 CatBoost, and LLM then mRMR the LLM-assisted leader on Home Credit logistic regression; only Stability 2024 logistic regression keeps its pair, pure LLM against RFE CatBoost, and with it its difference, and five of the six margins are wider than under the HO rule. On Stability 2024 CatBoost the development rule prefers the classical selector (0.8712 against 0.8662 on DEV), yet pure LLM is higher on HO by +0.028.

The two rules share five of their six LLM-assisted leaders, so a third and most adversarial pairing can be read from the same tables: the LLM-assisted leader chosen on development data against the classical leader chosen on the HO results. In five cases it is the pair of Table 3 and carries that table's difference and interval; on Home Credit logistic regression it pairs LLM then mRMR with mRMR, a difference of −0.027 reported as a point estimate, its interval not computed. The five-of-six pattern therefore holds under all three rules, and the LLM-assisted margins under the most adversarial rule are those of Table 3. The two rules are not, however, independent checks, and the reason is visible in the DEV and HO values that Table 4 prints side by side. Choosing a leader on development data can only lower or preserve its HO AUC relative to the HO-best selector of its family, and the rule costs the two families unequally. The DEV-chosen classical leader is the HO-best classical selector only on Stability 2024 logistic regression and sits below it by 0.001 to 0.017 in the other five cases, whereas the DEV-chosen LLM-assisted leader is the HO-best one in five cases and below it by 0.006 on Home Credit logistic regression. The development rule widens the margin because it costs the classical family more, not because the LLM-assisted leaders rise. The same asymmetry appears within each leader: from DEV mean to HO AUC the classical leaders of Table 4 fall by 0.008 to 0.022 in five cases and hold level on Stability 2024 logistic regression. The LLM-assisted leaders move by −0.010 to +0.012 and rise in three cases. The smaller development-to-holdout change of the LLM-assisted subsets is a finding in its own right, consistent with their lower selected-feature drift (Table 3), and it means that the agreement between the two leader rules is partly a consequence of that finding rather

than an independent corroboration of it.

The column "Full" is the HO AUC of the untuned full-feature reference, fitted on every candidate feature (on Stability 2024, the unscreened 1,959) under the same fixed configuration; it is context only and supports no claim about compact subsets against the full feature set (Section 5.3). Table 5 reports the HO AUC of every selector of the two families in every case, so that each leader margin can be read against the spread of its own family. On Home Credit the order of mRMR alone and LLM then mRMR depends on the backbone: mRMR on the whole candidate set is higher with logistic regression and LLM then mRMR with CatBoost (Table 5). The twelve headline subsets themselves, the full-DEV selections of the two leaders of each case, are listed in the supplementary material (Appendix D). Protected attributes (Section 4.4) appear in the subsets of both families. On Home Credit, `CODE_GENDER` holds rank 3 in the mRMR logistic-regression subset, rank 14 in the RFE CatBoost subset and rank 10 in the pure LLM CatBoost subset, and `NAME_FAMILY_STATUS` holds rank 9 in the pure LLM CatBoost subset and rank 39 in the RFE CatBoost subset; on Stability 2024 the two aggregates of the person-table sex field occupy ranks 1 and 6 of the CatBoost SHAP subset and no LLM-assisted subset contains one; the LendingClub v2 headline subsets contain no such field. Their presence is a property of the released universes rather than of either family, and the expert persona of the prompt did not exclude them unasked. Pure LLM screening is the LLM-assisted leader in four of the six cases and within 0.006 of it on Home Credit logistic regression; LLM then mRMR leads it by 0.037 on LendingClub v2 CatBoost. LLM then Boruta has the lowest AUC among the four pipelines in every case, which is not an artefact of short subsets: the Boruta stage reached its budget in 104 of its 108 fold and full-DEV refits, every full-DEV refit among them (Appendix F). On Stability 2024 the four LLM-assisted pipelines lie within 0.022 of one another with logistic regression and within 0.035 with CatBoost, pure LLM leading under both backbones (Table 5); resolved by month of decision, pure LLM has the higher AUC in every one of the nine months of the held-out window under both backbones (Appendix F).

### *6.2. Robustness of the headline comparison*

The five-of-six result of Table 3 could be an artefact of how the compared pair was chosen, of the encoding, of the composition of a holdout, of the tables a subset draws from, or of the particular language-model response behind it. Table 6 collects the twelve checks that address these readings, each with the alternative it rules out, its result and where its detail is reported. Under every check the LLM-assisted leader remains ahead in the same five cases; on Home Credit logistic regression the classical leader remains ahead under all three leader rules and under every re-ranking, and only the family cap, a control that handicaps the classical selector, brings the two families level there. The changes in AUC produced by re-ranking with obfuscated names, with descriptions rebuilt from lineage or with a repeated response all lie inside the materiality bar of 0.010 (Section 5.6), and the L1 selector run after the fact changes no leader. The detail behind the checks not reported elsewhere in this section, with Tables F.1, F.2 and F.3, is in Appendix F.

### *6.3. What drives the result*

Three readings of the headline result are tested here: that the ranking rests on recognising familiar feature names, that it rests on hand-written descriptions, and that it rests on drawing from a few source tables.

#### *6.3.1. Name obfuscation on Home Credit*

Table 7 reports the two obfuscated arms of Section 4.4 against the named pure LLM pipelines of Table 5, on the same HO observations. With the names replaced by identifiers and the definitions kept (arm A), the held-out AUC is unchanged, moving by $-0.0012$ with logistic regression and $+0.0042$ with CatBoost, both inside the materiality bar (Section 5.6). With the names kept and the definitions removed (arm B), the AUC falls under both backbones, by $-0.0290$ and $-0.0310$, three times the bar. The lists themselves change more than the AUC does. In every named Home Credit call the three external scores occupy ranks 1, 2 and 3, an order that nothing in their metadata can produce, since the three records are identical apart from the digit. In arm A they fall to ranks 10 to 18 in the five fold calls and out of the top three in the full-DEV call (Table 7); in arm B they sit at ranks 6, 7 and 8 in all six calls. Each arm's subset shares only part of its names with the named subset of the same partition and budget, the full-DEV overlaps lying at the low end of the fold ranges (Table 7). Across the five fold calls the arm A lists agree with one another more than the named lists do, and the arm B lists more still (pairwise Jaccard in the note to Table 7). The reading is the first of the three fixed before the run: the AUC holds without the names while the lists partly overlap, so the value of the ranking rests on the definitions, and the familiar names shaped the top of the named lists without carrying their predictive value. Section 6.3.2 adds that the definitions' contribution does not depend on their credit phrasing: with the names kept, descriptions rebuilt from lineage alone give the same held-out AUC within the materiality bar. Two limits apply. Obfuscating a name leaves a definition that may itself be recognisable, so the ablation weakens the name channel rather than removing it; and the comparison is descriptive, the differences of arm A being smaller than the fold-to-fold spread of a single arm, and covers Home Credit only, the dataset on which the exposure is largest (Section 7.6).

#### *6.3.2. Descriptions built from lineage*

Table 8 reports the lineage-only arm of Section 4.4 against the cached rankings, on the same HO observations. Held-out AUC moves by $-0.0064$ to $+0.0066$ and

Table 3: Primary held-out (HO) comparison. In each dataset–backbone case the LLM-assisted and the classical family leader, the selector with the highest observed HO AUC within its family, are compared on identical HO observations; both were chosen on the HO results, so the differences are descriptive and their intervals are not adjusted for that selection (Section 5.6). AUC is shown to four decimals; ΔAUC, LLM-assisted minus classical, and its paired 95% bootstrap interval are computed at full precision. "Full" is the HO AUC of the untuned full-feature reference with the same fixed configuration, context only (Section 5.3). The PSI columns give the mean DEV-versus-HO population stability index of the features selected by each leader's frozen full-DEV refit (Section 5.5); lower is less drift. Table 4 repeats the comparison with leaders chosen on development data.

| Case | LLM-assisted leader (AUC) | Classical leader (AUC) | Full | ΔAUC [95% CI] | PSILLM | PSIclassical |
|---|---|---|---|---|---|---|
| Home Credit, LR | Stable core + LLM fill (0.7489) | mRMR (0.7699) | 0.7715 | −0.021033 [−0.022211, −0.019780] | 0.0053 | 0.0085 |
| Home Credit, CatBoost | Pure LLM (0.7935) | RFE CatBoost (0.7814) | 0.7860 | +0.012024 [+0.009313, +0.014841] | 0.0045 | 0.0097 |
| LendingClub v2, LR | Pure LLM (0.7402) | IV then Boruta (0.7098) | 0.7109 | +0.030431 [+0.029359, +0.031514] | 0.0125 | 0.0129 |
| LendingClub v2, CatBoost | LLM then mRMR (0.7707) | IV then Boruta (0.7201) | 0.7230 | +0.050517 [+0.049712, +0.051315] | 0.0104 | 0.0125 |
| Stability 2024, LR | Pure LLM (0.8344) | RFE CatBoost (0.8214) | 0.7719 | +0.013006 [+0.010641, +0.015271] | 0.0174 | 0.0092 |
| Stability 2024, CatBoost | Pure LLM (0.8784) | CatBoost SHAP (0.8546) | 0.8245 | +0.023756 [+0.020961, +0.026611] | 0.0024 | 0.0189 |

Table 4: Development-rule held-out (HO) comparison with leaders chosen on development data. In each dataset–backbone case the LLM-assisted and the classical family leader are the selectors with the highest mean validation AUC over the five DEV folds within their family; the two are compared once on identical HO observations. Each leader is shown with its DEV mean and, after the semicolon, its HO AUC. ΔAUC, LLM-assisted minus classical, carries its paired 95% bootstrap interval and is read against the materiality bar (Section 5.6). The holdout plays no part in the choice of the compared methods, but the rule was specified after the results of Table 3 were known (Appendix E). The change from the DEV mean to the HO AUC of each leader is discussed in Section 6.1.

| Case | LLM-assisted leader (DEV; HO) | Classical leader (DEV; HO) | ΔAUC [95% CI] |
|---|---|---|---|
| Home Credit, LR | LLM then mRMR (0.7472; 0.7432) | IV then Boruta (0.7652; 0.7534) | −0.010179 [−0.012863, −0.007592] |
| Home Credit, CatBoost | Pure LLM (0.7917; 0.7935) | IV then Boruta (0.7770; 0.7688) | +0.024700 [+0.021656, +0.027770] |
| LendingClub v2, LR | Pure LLM (0.7436; 0.7402) | RFE CatBoost (0.7211; 0.6989) | +0.041353 [+0.040619, +0.042072] |
| LendingClub v2, CatBoost | LLM then mRMR (0.7811; 0.7707) | RFE CatBoost (0.7347; 0.7190) | +0.051702 [+0.050900, +0.052499] |
| Stability 2024, LR | Pure LLM (0.8258; 0.8344) | RFE CatBoost (0.8213; 0.8214) | +0.013006 [+0.010641, +0.015271] |
| Stability 2024, CatBoost | Pure LLM (0.8662; 0.8784) | RFE CatBoost (0.8712; 0.8500) | +0.028431 [+0.025711, +0.031121] |

every interval lies inside the materiality bar, the widest bound being +0.0086; by the rule of Section 5.6 none of these differences counts as a result in either direction. The five LLM-assisted leaders that exceed the classical leader in Table 3 are all higher with the rebuilt descriptions, so they would lead by +0.014 to +0.057 rather than by the margins of Table 3, every margin above the bar (point estimates, intervals not computed). On Home Credit logistic regression, the case the classical leader wins, stable core plus LLM fill moves by −0.0018 and stays below mRMR, a difference of −0.023 against −0.021 with the cached descriptions (Table 8). None of the six comparisons of Table 3 therefore depends on the phrasing of the descriptions, including the four on LendingClub v2 and Stability 2024, whose base definitions the first author wrote (Section 4.4). The headline tables keep the cached rankings, which were frozen before HO scoring; the lineage-only arm is a robustness check on them, not a replacement. On Stability 2024 both differences are smaller than the spreads of the ten repeated calls on the same partition (Table 10). As with the obfuscated arms, the subsets change more than the AUC does: the rebuilt full-DEV subsets share only part of their names with the cached ones (Overlap column of Table 8).

### *6.3.3. Source concentration*

The pure LLM subsets on Stability 2024 are drawn entirely from `static_0` under both budgets, and on Home Credit CatBoost 26 of the 40 features are bureau aggregates, whereas the classical leaders of those cases spread across six or seven source families (Appendix D). Whether

Table 5: Held-out (HO) ROC-AUC of every selector of the classical and LLM-assisted families in the six dataset–backbone cases ($K = 20$ for logistic regression, $K = 40$ for CatBoost). Bold marks the family leader in each column, the selector with the highest observed HO AUC within its family; the last row is the LLM-assisted leader minus the classical leader, the ΔAUC of Table 3 rounded to three decimals, which that table reports at full precision.

| | Home Credit | | LendingClub v2 | | Stability 2024 | |
|---|---|---|---|---|---|---|
| Selector | LR | CatBoost | LR | CatBoost | LR | CatBoost |
| *Classical family (nine selectors)* | | | | | | |
| Domain rules | 0.7230 | 0.7293 | 0.5525 | 0.6128 | 0.7416 | 0.7871 |
| Random $K$ | 0.6278 | 0.6854 | 0.6038 | 0.6718 | 0.7043 | 0.7516 |
| PCA | 0.6838 | 0.7091 | 0.6624 | 0.6787 | 0.7116 | 0.7770 |
| IV/WOE | 0.7381 | 0.7686 | 0.6821 | 0.7027 | 0.7844 | 0.8265 |
| mRMR | **0.7699** | 0.7599 | 0.6859 | 0.7020 | 0.7893 | 0.8330 |
| Boruta RF | 0.7397 | 0.7542 | 0.6557 | 0.6857 | 0.7693 | 0.8174 |
| IV then Boruta | 0.7534 | 0.7688 | **0.7098** | **0.7201** | 0.8030 | 0.8433 |
| RFE CatBoost | 0.7465 | **0.7814** | 0.6989 | 0.7190 | **0.8214** | 0.8500 |
| CatBoost SHAP | 0.7501 | 0.7795 | 0.6965 | 0.7151 | 0.8156 | **0.8546** |
| *LLM-assisted family (four pipelines)* | | | | | | |
| Pure LLM | 0.7436 | **0.7935** | **0.7402** | 0.7335 | **0.8344** | **0.8784** |
| LLM then mRMR | 0.7432 | 0.7758 | 0.7179 | **0.7707** | 0.8159 | 0.8554 |
| LLM then Boruta | 0.7337 | 0.7657 | 0.6793 | 0.7002 | 0.8123 | 0.8435 |
| Stable core + LLM fill | **0.7489** | 0.7809 | 0.7048 | 0.7277 | 0.8261 | 0.8592 |
| LLM-assisted leader − classical leader | −0.021 | +0.012 | +0.030 | +0.051 | +0.013 | +0.024 |

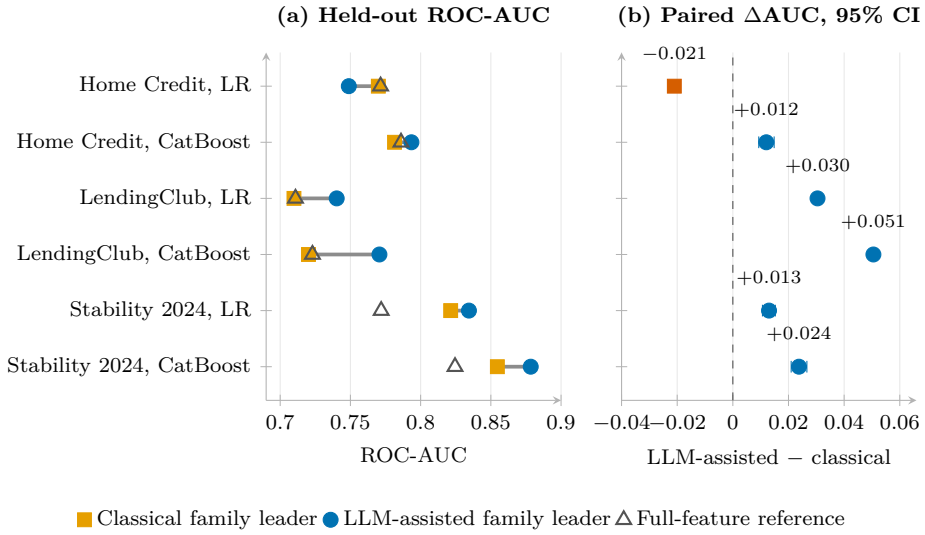



Figure 3: Primary held-out (HO) comparison. (a) ROC-AUC of the classical family leader, the LLM-assisted family leader (the selector with the highest observed HO AUC within its family) and the untuned full-feature reference in each dataset–backbone case; the grey bar joins the two selectors being compared. (b) Paired difference, LLM-assisted minus classical, with its 95% confidence interval; positive values favour LLM-assisted selection and the red square marks the one case in which the classical selector is higher. The comparison is descriptive, its paired bootstrap intervals not adjusted for the selection of the compared methods on the same HO results (Section 5.6); Table 4 gives the development-rule comparison. Plot code generated with AI assistance from the study's result files (see the declaration on generative AI).

a rule concentrating on the same tables would do as well is tested by the two diversity controls of Section 4.3: capping the classical leader per source family lowers it in all six cases and leaves it below the LLM-assisted leader in five, and confining it to the two static tables on Stability 2024 leaves it 0.060 and 0.058 below pure LLM over the same pool (Table F.2; Appendix F). Source concentration alone therefore does not reproduce the Stability 2024 margins.

### *6.4. Stability, drift and secondary metrics*

#### *6.4.1. Fold-to-fold stability and response sampling*

Table 9 and Figure 4 report fold-to-fold reproducibility (RQ2). Among independently refitted classical selectors, Boruta RF on LendingClub logistic regression is the most reproducible selector in the study (Nogueira 0.9588, Jaccard 0.9273); IV/WOE is the most reproducible classical selector in four of the six cases and mRMR on LendingClub CatBoost. Stability is lowest on Stability 2024, whose candidate set of 1,068 features is by far the largest (Nogueira

Table 6: Robustness of the headline comparison. Each row is one check on the five-of-six result of Table 3, the alternative reading it rules out, its result and where its detail is reported; differences are paired HO ΔAUC, LLM-assisted minus classical unless stated, read against the materiality bar of 0.010 (Section 5.6). HC LR is Home Credit with logistic regression.

| Check | What it rules out | Result | Headline holds? | Source |
|---|---|---|---|---|
| Leaders chosen on DEV folds | Pair chosen on HO results | LLM-assisted leader higher in 5 of 6, +0.013 to +0.052; classical leads HC LR by 0.010 | Yes | Table 4 |
| DEV-chosen LLM-assisted vs HO-best classical | A favourable pairing | Held-out-rule margins in five cases; HC LR −0.027 | Yes | Section 6.1 |
| Native categorical handling (CatBoost) | One-hot artefact | AUC moves by at most 0.005; ordering unchanged in all three cases | Yes | Appendix F |
| Per-family cap on the classical leader | Source concentration | Classical leader falls by 0.009 to 0.064; stays 0.021 to 0.077 below in 5 of 6; on HC LR the families end level within the bar (−0.003) | Yes | Table F.2 |
| Depth-0 restriction (Stability 2024) | Same-table pool | Classical leader 0.060 (LR) and 0.058 (CatBoost) below pure LLM | Yes | Table F.2 |
| Matured cohort (LendingClub) | Resolution filter | +0.018 (LR), +0.011 (CatBoost) | Yes | Table F.1 |
| Term split (LendingClub) | Term mix | 36 months +0.022 / +0.012; 60 months +0.014 / +0.010 | Yes | Table F.1 |
| Monthly breakdown (Stability 2024) | One favourable period | Pure LLM higher in all nine months under both backbones | Yes | Appendix F |
| Name obfuscation (Home Credit) | Name recall | −0.0012 (LR), +0.0042 (CatBoost) | Yes | Table 7 |
| Descriptions rebuilt from lineage | Hand-written phrasing | Held-out AUC moves by −0.0064 to +0.0066 | Yes | Table 8 |
| Ten repeated LLM calls (Stability 2024) | A favourable response | Spread 0.0095 (LR) and 0.0092 (CatBoost); direction unchanged in all 20 refits | Yes | Section 6.4.1 |
| L1-penalised selector | Missing standard baseline | Below the classical leader in every case | Yes; changes no leader | Appendix F |

Table 7: Name-obfuscation ablation on Home Credit. Held-out AUC of the pure LLM pipeline when the six ranking calls are repeated with the feature names replaced by seeded identifiers (arm A, definitions kept) or with the definitions removed (arm B, names kept), against the named cached rankings; same HO observations, same fixed backbones. Fold range is the span of the five fold-model AUCs. Overlap is the number of names the arm's subset shares with the named subset of the same partition and budget: for the full-DEV call (Appendix D) and, as a range and mean, over the five fold calls. Ranks of the three external scores are given for the full-DEV call.

| Arm | Model | $K$ | HO AUC | Named | Δ | Fold range | Overlap, full DEV | Overlap, folds |
|---|---|---|---|---|---|---|---|---|
| A, names removed | LR | 20 | 0.7424 | 0.7436 | −0.0012 | 0.721–0.741 | 11/20 | 12–18 (mean 14.2) |
| A, names removed | CatBoost | 40 | 0.7977 | 0.7935 | +0.0042 | 0.751–0.785 | 17/40 | 15–28 (mean 23.2) |
| B, definitions removed | LR | 20 | 0.7146 | 0.7436 | −0.0290 | 0.706–0.735 | 14/20 | 14–19 (mean 16.2) |
| B, definitions removed | CatBoost | 40 | 0.7625 | 0.7935 | −0.0310 | 0.731–0.753 | 24/40 | 26–33 (mean 28.8) |

Ranks of `EXT_SOURCE_1/2/3` in the full-DEV call: named 1, 2, 3; arm A 12, 57, 13; arm B 6, 7, 8. Pairwise Jaccard among the five fold lists at $K = 20$: arm A 0.43 to 1.00 (mean 0.60), arm B 0.67 to 1.00 (mean 0.78), named lists 0.25 to 0.67 (mean 0.41).

about 0.78, Jaccard about 0.66). The pure LLM lists were requested separately per fold partition rather than refitted, and Table 9 reports their fold agreement over the candidate set on Home Credit and LendingClub, where five fold lists exist per case; the cached rankings all differ, the six of 100 names on Home Credit and all 24 on LendingClub. Their agreement is among the lowest in the table, with LLM then Boruta: Nogueira 0.41 to 0.63 and mean pairwise Jaccard 0.31 to 0.48, with the fold-pair minimum and maximum of each list in Table 9. The fold-pair spreads of the pure LLM lists, 0.14 to 0.42 wide, are comparable in width to those of the most reproducible classical selectors, 0.18 to 0.42, so the lists differ at a lower level of agreement rather than more erratically. The ordering holds pair by pair: on Home Credit and LendingClub the least similar fold pair of the most reproducible classical selector is at least as similar as the most similar pair of the pure LLM list of the same case (Min and Max columns of Table 9), so the difference between the two families does not rest on a single fold split. The names shared by all five fold lists of a case and the names the five lists span together are given in Table 11.

Table 8: Descriptions built from lineage. Held-out AUC when every description is rebuilt from lineage alone and the ranking calls are repeated (Mechanical), against the cached rankings of Table 5 (Cached), for pure LLM in the six cases and for the LLM-assisted leader of the two cases in which it is not pure LLM (stable core plus LLM fill on Home Credit LR, LLM then mRMR on LendingClub v2 CatBoost); same HO observations, same frozen pipelines. $\Delta$ is mechanical minus cached with its paired 95% bootstrap interval, computed before rounding; like the intervals of Table 3 it holds both rankings fixed, and it is read against the materiality bar of 0.010 (Section 5.6). Overlap is the number of names of the full-DEV subset shared with the cached subset; for LLM then mRMR it counts budget slots, which include one-hot indicators (Section 4.2).

| Case | Pipeline | $K$ | Mechanical | Cached | $\Delta$ [95% CI] | Overlap |
|---|---|---|---|---|---|---|
| Home Credit, LR | Pure LLM | 20 | 0.7398 | 0.7436 | −0.0039 [−0.0053, −0.0017] | 12/20 |
| Home Credit, LR | Stable core + LLM fill | 20 | 0.7471 | 0.7489 | −0.0018 [−0.0029, −0.0007] | 17/20 |
| Home Credit, CatBoost | Pure LLM | 40 | 0.7953 | 0.7935 | +0.0019 [+0.0013, +0.0024] | 24/40 |
| LendingClub v2, LR | Pure LLM | 20 | 0.7444 | 0.7402 | +0.0041 [+0.0027, +0.0063] | 12/20 |
| LendingClub v2, CatBoost | Pure LLM | 40 | 0.7271 | 0.7335 | −0.0064 [−0.0072, −0.0057] | 24/40 |
| LendingClub v2, CatBoost | LLM then mRMR | 40 | 0.7773 | 0.7707 | +0.0066 [+0.0056, +0.0074] | 26/40 |
| Stability 2024, LR | Pure LLM | 20 | 0.8396 | 0.8344 | +0.0052 [+0.0034, +0.0086] | 15/20 |
| Stability 2024, CatBoost | Pure LLM | 40 | 0.8791 | 0.8784 | +0.0007 [−0.0005, +0.0008] | 30/40 |

On Stability 2024 one ranking served every fold, so its fold agreement is 1 by construction and is not reported; in its place the sampling of the model at a fixed prompt was isolated directly by repeating the full-DEV call ten times with the same prompt, snapshot, temperature 0 and the same 1,068 candidate records. All ten responses passed the same strict validation as the cached one, and the ten rankings of 100 names are all distinct. Over that candidate set the ten lists reach Nogueira 0.8052 at $K = 20$ and 0.7766 at $K = 40$ (Table 11), level with the most reproducible classical selector of the same case over the same universe (0.7860 and 0.7793); the mean pairwise Jaccard of the ten lists at $K = 20$, 0.6908, exceeds that selector's fold mean of 0.6666 and lies inside its fold-pair range (Table 9). The pure LLM rows of Table 9 for Home Credit and LendingClub measure the same quantity, the candidate list being fixed there too (Section 4.4), and are much lower; because the chance correction depends on $d$ (Section 3.4) the three are not on a common scale and the comparison between datasets is not made here. Carrying each response through the frozen downstream pipeline gives the held-out AUC ranges, medians and standard deviations of Table 10; the cached values of Table 3 lie above four and above seven of the ten repeats, inside the body of the distribution rather than at its top. The margin over the classical leader stays positive in all twenty refits (Table 10), so no repeat reverses the direction of the comparison; two of the ten logistic-regression repeats fall below the materiality bar of 0.010 (Section 5.6), by 0.0014 and 0.0003, while every CatBoost repeat clears it. Both spreads, 0.0095 at $K = 20$ and 0.0092 at $K = 40$, are themselves inside that bar, so on this dataset the choice of cached response is immaterial at both budgets by the study's own standard. These twenty refits vary the response only and are not a confidence interval; they are reported beside, not inside, the paired bootstrap intervals of Section 5.6. The same quantity was measured on Home Credit and LendingClub, where five semantic rankings per dataset were cached, one per fold partition, over the identical candidate list; each ranking's top $K$ was carried through the frozen full-DEV pipeline and scored once on HO, twenty refits in all. Held-out AUC spans at most 0.0075 across the five responses in the four Home Credit and LendingClub cases (Table 10), every spread inside the materiality bar of 0.010 and narrower than the two spreads measured on Stability 2024. Response sampling at a fixed prompt is therefore immaterial at the held-out AUC level in all three datasets.

The hybrids' final subsets depend on both the per-fold semantic ranking and the fitted statistical stage. Stable core plus LLM fill is the more reproducible hybrid in every case, reaching Nogueira 0.69 to 0.76 (Table 9). Its Jaccard lies between 0.57 and 0.63. It sits below the most reproducible classical selector of the same case by 0.03 to 0.25 in Nogueira and above pure LLM by 0.08 to 0.28. Its fold pairs reach the classical range: in all six cases its most similar pair is at or above the least similar pair of the most reproducible classical selector (Min and Max columns of Table 9), so the construction recovers classical-level agreement on part of the fold pairs. LLM then Boruta lies between 0.42 and 0.63 in all six cases. On Home Credit and LendingClub it is level with the pure LLM lists it refits on, within 0.04 in three of the four cases and 0.11 below on Home Credit logistic regression; on Stability 2024, where one semantic ranking served every fold, its 0.49 (logistic regression) and 0.55 (CatBoost) are produced by the Boruta refit alone. Refitting Boruta on the semantic candidates therefore does not recover the reproducibility that the ranking loses: with the ranking held fixed it leaves the subsets 0.16 to 0.27 below the stable core construction of the same case (Table 9). It leaves them 0.23 to 0.30 below the most reproducible classical selector. The stable core construction recovers part of what the ranking loses, and the most reproducible classical selectors exceed pure LLM by 0.28 to 0.43 in Nogueira. In Figure 4 every point lies below the identity line, because over a candidate set of 373 to 1,068 features the chance correction of Eq. (8) is small and $\hat{\Phi}$ exceeds the uncorrected Jaccard; the correction grows as the universe shrinks, so for the same fold subsets a smaller universe gives a lower $\hat{\Phi}$, which is why the statistic is compared only within a dataset.

Table 9: Fold-to-fold reproducibility across the five development folds: the most reproducible independently refitted classical selector in each case (Nogueira stability as primary ordering, mean pairwise Jaccard as tie-breaker), the two LLM hybrids whose final subset is fixed by a refitted statistical stage, and the pure LLM lists cached per fold on Home Credit and LendingClub. Jaccard is the mean pairwise Jaccard over the ten fold pairs; Min and Max are its minimum and maximum over the same pairs. $d$ is the Nogueira universe.

| Case | Model | Selector | $K$ | Nogueira | Jaccard | Min | Max[b] |
|---|---|---|---|---|---|---|---|
| *Refitted classical selectors (d = the candidate set: 373, 675 and 1,068)* | | | | | | | |
| Home Credit | LR | IV/WOE | 20 | 0.8838 | 0.8053 | 0.667 | 0.905 |
| Home Credit | CatBoost | IV/WOE | 40 | 0.8403 | 0.7538 | 0.667 | 0.905 |
| LendingClub v2 | LR | Boruta RF | 20 | 0.9588 | 0.9273 | 0.818 | 1.000 |
| LendingClub v2 | CatBoost | mRMR | 40 | 0.9070 | 0.8462 | 0.702 | 1.000 |
| Stability 2024 | LR | IV/WOE | 20 | 0.7860 | 0.6666 | 0.481 | 0.905 |
| Stability 2024 | CatBoost | IV/WOE | 40 | 0.7793 | 0.6582 | 0.509 | 0.905 |
| *LLM hybrids with a refitted statistical stage (d = the candidate set: 373, 675 and 1,068)* | | | | | | | |
| Home Credit | LR | Stable core + LLM fill | 20 | 0.7411 | 0.6110 | 0.481 | 0.818 |
| Home Credit | CatBoost | Stable core + LLM fill | 40 | 0.6920 | 0.5725 | 0.455 | 0.739 |
| LendingClub v2 | LR | Stable core + LLM fill | 20 | 0.7115 | 0.5783 | 0.379 | 0.818 |
| LendingClub v2 | CatBoost | Stable core + LLM fill | 40 | 0.7130 | 0.5842 | 0.404 | 0.778 |
| Stability 2024 | LR | Stable core + LLM fill | 20 | 0.7554 | 0.6259 | 0.429 | 0.905 |
| Stability 2024 | CatBoost | Stable core + LLM fill | 40 | 0.7091 | 0.5727 | 0.379 | 0.778 |
| Home Credit | LR | LLM then Boruta | 20 | 0.4294 | 0.3031 | 0.212 | 0.429 |
| Home Credit | CatBoost | LLM then Boruta | 40 | 0.4204 | 0.3247 | 0.194 | 0.509 |
| LendingClub v2 | LR | LLM then Boruta | 20 | 0.5414 | 0.4103 | 0.143 | 0.739 |
| LendingClub v2 | CatBoost | LLM then Boruta | 40 | 0.6333 | 0.4920 | 0.379 | 0.702 |
| Stability 2024 | LR | LLM then Boruta | 20 | 0.4876 | 0.3682 | – | – |
| Stability 2024 | CatBoost | LLM then Boruta | 40 | 0.5528 | 0.4193 | – | – |
| *Pure LLM, cached per-fold lists (d = the candidate set: 373 and 675)*[a] | | | | | | | |
| Home Credit | LR | Pure LLM | 20 | 0.5403 | 0.4076 | 0.250 | 0.667 |
| Home Credit | CatBoost | Pure LLM | 40 | 0.4092 | 0.3149 | 0.231 | 0.509 |
| LendingClub v2 | LR | Pure LLM | 20 | 0.5517 | 0.4011 | 0.250 | 0.600 |
| LendingClub v2 | CatBoost | Pure LLM | 40 | 0.6280 | 0.4828 | 0.404 | 0.538 |

[a] Lists requested separately for each fold partition, not refitted; because the candidate set is the same in every call, a fold-to-fold difference reflects the sampling of the language model at a fixed prompt. The Nogueira universe is the same screened candidate set as for every other row of the dataset, 373 features on Home Credit and 675 on LendingClub, so the pure LLM rows are on a common scale with the classical and hybrid rows. Stability 2024 is absent because one ranking served every fold, so its fold agreement is 1 by construction.
[b] Minimum and maximum pairwise Jaccard over the ten fold pairs, from the same fold lists as the mean; not reported for the two Stability 2024 LLM then Boruta rows.

Table 10: Response sampling at a fixed prompt. Each row carries one cached semantic ranking's top $K$ through the frozen full-DEV pipeline to a single HO scoring run; the draws vary the cached response only and are not a confidence interval. Stability 2024 uses the ten repeated full-DEV calls of Section 6.4.1; Home Credit and LendingClub use the five rankings cached per fold partition over the identical candidate list. Spread is the range of held-out AUC over the draws, read against the materiality bar of 0.010 (Section 5.6); Table 11 gives the agreement among the draws.

| Case | Model | $K$ | Draws | HO AUC range | Spread | Median | s.d. | Margin over classical leader |
|---|---|---|---|---|---|---|---|---|
| Home Credit | LR | 20 | 5 | 0.7358–0.7410 | 0.0053 | – | – | – |
| Home Credit | CatBoost | 40 | 5 | 0.7926–0.7998 | 0.0072 | – | – | – |
| LendingClub v2 | LR | 20 | 5 | 0.7106–0.7181 | 0.0075 | – | – | – |
| LendingClub v2 | CatBoost | 40 | 5 | 0.7324–0.7362 | 0.0038 | – | – | – |
| Stability 2024 | LR | 20 | 10 | 0.8300–0.8395 | 0.0095 | 0.8350 | 0.0030 | +0.0086 to +0.0181 (RFE CatBoost, 0.8214) |
| Stability 2024 | CatBoost | 40 | 10 | 0.8740–0.8832 | 0.0092 | 0.8764 | 0.0030 | +0.0194 to +0.0286 (CatBoost SHAP, 0.8546) |

Every spread lies inside the materiality bar, so the choice of cached response is immaterial at the held-out AUC level in all three datasets by the study's own standard. Median, standard deviation and the margin over the classical leader of the case (its HO AUC in parentheses) are reported for the ten Stability 2024 repeats.

### *6.4.2. Drift and secondary metrics*

Table 12 reports the best held-out (HO) value of each secondary metric in each dataset with the pipeline attaining it; the leaders are not always the same pipeline, so the values describe the frontier reached by the selector family rather than one model. The best KS is attained by an LLM-assisted CatBoost pipeline in all three datasets, the best top-decile concentration by pure LLM in all three, and the best Brier score by an LLM-assisted CatBoost pipeline in each dataset. The best Brier score is the lowest among the evaluated selectors and not evidence of calibration, because both backbones use balanced class weights and no

Table 11: Agreement among the cached semantic responses of Table 10: Nogueira stability and mean pairwise Jaccard of the draws over the screened candidate set ($d = 373$, 675 and 1,068; the Home Credit and LendingClub rows are the pure LLM rows of Table 9), the number of the $K$ names present in every draw, and the number of distinct names the draws span together. Because the chance correction depends on $d$, values are not compared across datasets.

| Case | Model | $K$ | Draws | Nogueira | Jaccard | Names in all draws | Distinct names |
|---|---|---|---|---|---|---|---|
| Home Credit | LR | 20 | 5 | 0.5403 | 0.4076 | 6 | 42 |
| Home Credit | CatBoost | 40 | 5 | 0.4092 | 0.3149 | 8 | 98 |
| LendingClub v2 | LR | 20 | 5 | 0.5517 | 0.4011 | 4 | 39 |
| LendingClub v2 | CatBoost | 40 | 5 | 0.6280 | 0.4828 | 14 | 71 |
| Stability 2024 | LR | 20 | 10 | 0.8052 | 0.6908 | 11 | 30 |
| Stability 2024 | CatBoost | 40 | 10 | 0.7766 | 0.6506 | 22 | 69 |

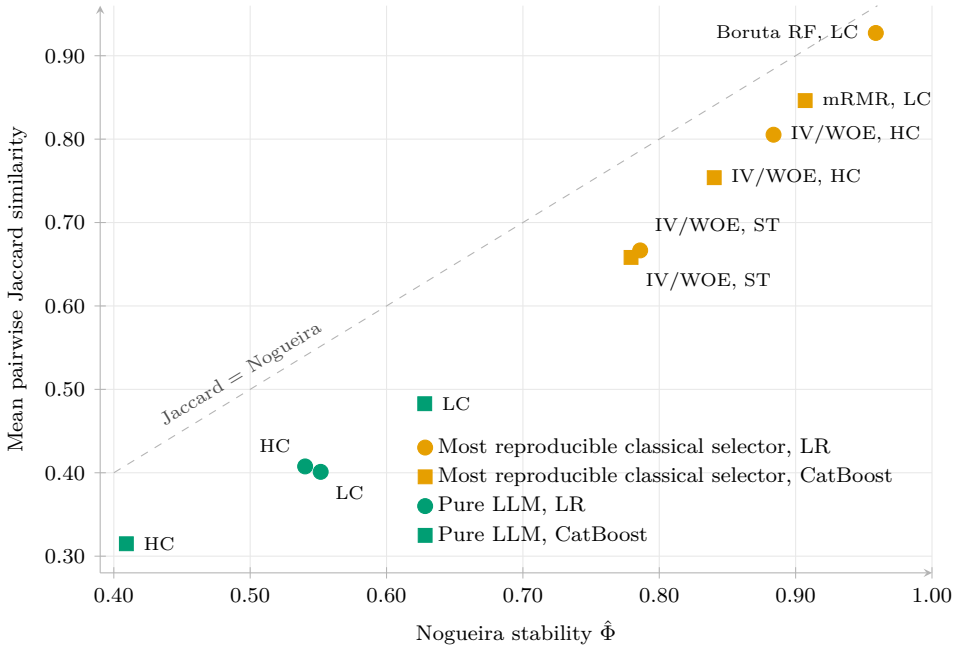



Figure 4: Reproducibility landscape of fold-level selections. Orange: most reproducible refitted classical selector in each dataset–backbone case; green: pure LLM lists cached per fold on Home Credit and LendingClub. Circles are logistic regression ($K = 20$), squares CatBoost ($K = 40$). HC = Home Credit, LC = LendingClub v2, ST = Stability 2024. Higher is more reproducible on both axes; the dashed line marks equality of the two statistics. The Nogueira universe is the screened candidate set of the dataset for every point shown, so all points are on a common scale within a dataset; because the chance correction depends on $d$, values are not compared across datasets. Plot code generated with AI assistance from the study's result files (see the declaration on generative AI).

recalibration (Section 5.5). Feature drift is reported for the twelve headline pipelines of Table 3, as the mean, median and maximum DEV-versus-HO PSI of the features each leader's frozen full-DEV refit selected. All values are small: mean selected-feature PSI runs from 0.0024 to 0.0189 across the twelve subsets, and no feature of any subset exceeds a PSI of 0.115. The LLM-assisted leader has the lower mean drift than the classical leader of the same case in five of the six cases and the lower median in all six (Table 3); the exception is Stability 2024 logistic regression, where pure LLM's mean exceeds RFE CatBoost's (0.0174 against 0.0092). Its median, 0.0023 against 0.0042, is still the lower. Every one of the twelve means sits an order of magnitude below the conventional action threshold of 0.10, so the selected subsets of both families would pass a drift review unchanged; the comparison between the families therefore reports a measurable difference in the composition of the subsets, consistent with the prompt's instruction to favour durable inputs, and not an operational advantage in drift. The median feature PSI of the pure LLM CatBoost subset on Home Credit is 0.0035, and on Stability 2024 CatBoost 0.0001, meaning that for the typical selected variable almost no distribution difference was detected under the implemented states, binning and precision. The largest single-feature drift in a headline subset is 0.114, a tax-registry row count in the CatBoost SHAP subset on Stability 2024, followed by 0.086 for a count of instalments without delinquency in the pure LLM subset of the same case and 0.058 for an instalment-timing variance in the RFE CatBoost subset on Home Credit; on Home Credit, where the held-out population is shifted in prior-application recency, the most drifted selected feature of every leader is a bureau, instalment or credit-card field rather than a previous-application recency field. Lower

drift within the selected inputs is reported as a property of the subsets and not as an explanation of the AUC ordering (Section 7.3).

### 6.5. Scalability and cost

One downstream saving from a reduced candidate pool was measured under matched conditions: on Stability 2024 fold 1, on the same machine, the mRMR stage took 522 seconds over the 1,959-feature universe and 21 seconds over a 100-feature pool, a factor of 25 for a pool twenty times smaller. This is the one runtime measurement of the study made under matched conditions and the only one on which a statement about scale rests. A pool of that size is what the semantic stage hands to the statistical stage of LLM then mRMR under CatBoost (Section 4.5), and that stage reads the variable book and no data rows, so its own cost depends on the number of candidate records and not on the number of decisions.

Table 13 sets out what each stage of the pipeline reads, how often it is recomputed and what its cost grows with, as these follow from the method definitions of Sections 4.3, 4.4 and 4.5; the table describes the design and is not a measurement. The semantic ranking is the only stage whose input is the variable book rather than the data rows: it was requested once per partition on Home Credit and LendingClub and once on Stability 2024, and the cached response served every later refit. Every other selection stage, and every model fit, is recomputed in every fold and again in the full-DEV refit, over all the rows of its partition.

On Home Credit and LendingClub, 72 logical screening requests were served by 30 provider ranking calls, 6 on Home Credit and 24 on LendingClub, the remainder being cache hits. The 30 calls consumed 806,757 input and 56,495 output tokens, for a direct API cost between USD 0.171 and USD 0.413 at the GPT-4.1 mini list prices of USD 0.40 per million input tokens, 0.10 per million cached input tokens and 1.60 per million output tokens; the bounds correspond to all input charged at the cached rate and at the uncached rate. The Stability 2024 ranking call is not included, because its token usage was not recorded.

The saved runs of the final protocol were executed on the CPU of one machine with 16 logical (10 physical) cores and 39.6 GiB of RAM, with four estimator threads and no accelerator; peak process memory was 7.4 to 10.7 GB for the Home Credit selectors and full-feature references, 19.1 to 29.0 GB for their LendingClub counterparts and 27.0 to 29.0 GB for their Stability 2024 counterparts. Wall-clock times of the pipelines were measured under unequal caching and execution conditions and support no scalability claim; they are reported for Home Credit and LendingClub in Appendix F.

## 7. Discussion

### 7.1. Main Findings

RQ1 asked whether an LLM pre-screened subset can match or exceed classically selected subsets on held-out data under a fixed budget. The answer is conditional: in five of six dataset–backbone cases the LLM-assisted family leader exceeds the classical family leader, whether the leaders are chosen on the HO results, in which case the comparison is descriptive, on the development folds, or under the most adversarial pairing of the two (Section 6.1, Table 4); on Home Credit logistic regression the classical leader is higher under all three rules, mutual-information mRMR by 0.021 and 0.027 and IV then Boruta by 0.010. The three rules agree partly for one reason, that the LLM-assisted subsets lose less AUC from development to holdout than the classical ones, so their agreement is not three independent confirmations (Section 6.1). The pattern does not depend on comparing family leaders: by mean HO AUC over the selectors that are not declared floors, the LLM-assisted family is ahead in the same five cases and behind on Home Credit logistic regression (Section 6.1).

The leading LLM-assisted method is pure LLM in four cases, LLM then mRMR in one (LendingClub v2 CatBoost) and stable core plus LLM fill in the Home Credit logistic-regression case that the classical selector wins. Within these comparisons the semantic ranking therefore most often had the higher HO AUC when used directly; statistical refinement of the semantic pool led in one case, and on Home Credit CatBoost the refined pipeline (LLM then mRMR) had a lower held-out AUC than pure LLM (Table 5). Direct use of the semantic ranking and statistical refinement are alternatives whose observed value differed by dataset and backbone. The results establish neither arrangement as generally preferable, nor LLM pre-screening as a replacement for statistical selection. The choice between them for a deployed pipeline is a development-stage decision, to be made on development data rather than on the holdout used to report final performance, which is how the leaders of Table 3 were identified and why Table 4 repeats the comparison with leaders chosen on development data. The full-feature reference is untuned and context only (Section 5.3).

On reproducibility (RQ2), refitted classical selectors reach Nogueira stability between 0.78 and 0.96, with information value and Boruta the most reproducible. The pure LLM lists of the discrimination-leading pipelines differ from fold to fold, Nogueira 0.41 to 0.63 over the same candidate set (Section 6.4.1), and the best-discriminating selector in this study is, with LLM then Boruta, the least reproducible one in Table 9, and its reported AUC is that of the particular cached responses. Ten repeated calls on the Stability 2024 partition bound how much that last point costs there: the ranking varies (Table 11) but the held-out AUC moves by 0.0095 and 0.0092, inside the materiality bar, and the direction of the comparison holds in every repeat (Section 6.4.1). A diversity control separates

Table 12: Best held-out (HO) secondary metric in each dataset, with the selector and backbone attaining it. KS is the maximum separation over HO thresholds; Brier is lower-is-better and, with balanced class weights and no recalibration, does not indicate calibration (Section 5.5); Capture@10 is the share of events in the highest-scored decile, with Lift@10 in parentheses.

| Dataset | KS | Brier score | Capture@10 (Lift@10) |
|---|---|---|---|
| Home Credit | 0.4264<br>Pure LLM, CatBoost | 0.1234<br>Pure LLM, CatBoost | 0.3630 (3.63)<br>Pure LLM, CatBoost |
| LendingClub v2 | 0.3898<br>LLM then mRMR, CatBoost | 0.1824<br>LLM then mRMR, CatBoost | 0.2751 (2.75)<br>Pure LLM, LR |
| Stability 2024 | 0.5559<br>Pure LLM, CatBoost | 0.1592<br>Pure LLM, CatBoost | 0.4671 (4.67)<br>Pure LLM, CatBoost |

Table 13: Cost structure of the pipeline stages, as it follows from the method definitions of Sections 4.3, 4.4 and 4.5; a description of the design, not a measurement. "Recomputed" states how often the stage runs in the protocol; "Cost grows with" names the quantities its work scales with. IV then Boruta, LLM then mRMR and LLM then Boruta combine the rows of their two stages, the second running over the pool the first leaves.

| Stage | Reads | Recomputed | Cost grows with |
|---|---|---|---|
| Semantic ranking (LLM) | Variable book, one record per candidate; no data rows | Once per partition on Home Credit (per partition and budget on LendingClub); once on Stability 2024; cached thereafter | Number of candidate records; not the number of decisions |
| Missingness screen | Feature values of the earliest training partition | Once per dataset | Rows × candidates |
| IV/WOE | Feature values and labels | Every fold and the full-DEV refit | Rows × candidates |
| mRMR | Feature values and labels | Every fold and the full-DEV refit | One relevance estimate per candidate and up to $K$ redundancy estimates per candidate, each over all rows |
| Boruta RF | Feature values, one shuffled shadow copy per feature, and labels | Every fold and the full-DEV refit | Random-forest fits over all rows and twice the number of candidates, repeated for the confirmation test |
| RFE CatBoost | Feature values and labels | Every fold and the full-DEV refit | Repeated boosted fits over all rows as the pool shrinks by 20% per step |
| CatBoost SHAP | Feature values and labels | Every fold and the full-DEV refit | One boosted fit, plus SHAP values over a 10,000-row sample |
| Stable core + LLM fill | Feature values and labels, plus the semantic ranking | Every fold and the full-DEV refit | Five bootstrap runs of the RF/correlation filter |
| Model fitting | The selected subset | Every fold and the full-DEV refit | Rows × selected columns |

that result from source concentration: a classical selector confined to the two static tables the semantic ranking drew from on Stability 2024 stays 0.060 and 0.058 below pure LLM over the same pool, and a per-family cap leaves the classical leader below its LLM-assisted counterpart in five of the six cases (Table F.2), so the margins are not reproduced by concentrating on the same tables.

The obfuscation ablation (Section 6.3.1) addresses the reading of the contribution that the study could not otherwise exclude, recognition of familiar feature names rather than semantic pre-screening. On Home Credit, the dataset whose variable book is most widely documented, replacing every name by a seeded identifier leaves the held-out AUC of the pure LLM pipeline within 0.005 of the named result under both backbones, while removing the definitions and keeping the names costs about 0.03. The ranking's value therefore does not rest on which names appear, nor, by the lineage-only arm (Section 6.3.2), on hand-written descriptions: with the names kept, descriptions generated from lineage serve as well in all three datasets. The names did leave a trace: in every named call the three external scores are ranked 1, 2, 3, an order the identical metadata of the three cannot produce, and it disappears when the names do. That ordering is the clearest evidence of recall in the study, and the ablation shows that it did not carry the result. The test covers one dataset and weakens rather than removes the name channel, since a definition can itself be recognisable; its scope is stated in Section 7.6.

### *7.2. Supplementary Experiment: Classical Rank Voting*

The classical rank-voting family (Appendix B) asked whether aggregating several classical rankings over a wider

candidate pool raises held-out discrimination over a single classical filter. Its twelve comparisons, three pool sizes in each of the four Home Credit and LendingClub cases, were fixed in a protocol file before any held-out result of the family was computed, and every one is positive, from +0.0020 to +0.0129 AUC, with paired bootstrap intervals that exclude zero and Holm-adjusted paired DeLong tests significant within each family (Table B.1). This is the one confirmatory result in the paper: every other held-out comparison, including the six headline comparisons under each leader rule, is reported as an effect size with an interval and no test (Section 5.6). The family contains no semantic voter, so it shows that rank voting can raise held-out AUC on both datasets and does not bear on the semantic pipelines; the voted subsets were also less reproducible across folds than the reference filter.

### *7.3. Predictive Performance vs Stability*

Three trade-offs emerge. The LLM-assisted pipeline that leads discrimination on LendingClub rests on cached per-fold rankings whose fold agreement is among the lowest in Table 9: calls with identical prompt, model and temperature returned different orderings, and the pure LLM lists reach Nogueira 0.41 to 0.63 over the same candidate set on which the most reproducible classical selectors reach at least 0.78 (Table 9), with LLM then Boruta no more reproducible than the lists it refits on (Section 6.4.1). Their reproducibility as deployed selectors therefore rests on the cached responses, not on the determinism of the language model under a fixed snapshot and prompt. Repeating one call ten times on Stability 2024 places that response-sampling component at Nogueira 0.8052 and 0.7766 over the same candidate set (Table 11). It moves held-out AUC by 0.0095 and 0.0092, both inside the materiality bar of Section 5.6, with the direction of the comparison unchanged in all twenty refits (Section 6.4.1); on that dataset caching the response therefore fixes a choice that was not, in the event, a material one. The most reproducible classical selectors (IV/WOE, Boruta RF) are not the classical leaders in discrimination (RFE CatBoost, IV then Boruta, mRMR), which are wrappers or greedy procedures; greater sensitivity to the training sample is a possible explanation. The stable-core construction is the device intended to raise the reproducibility of a semantically screened subset; the stable-core hybrid is the LLM-assisted family leader on Home Credit logistic regression, although it remains below mRMR there, and its fold stability, 0.08 to 0.28 above pure LLM in Nogueira, is reported in Table 9.

CatBoost exceeds logistic regression in every headline case. The two backbones differ in both model class and feature budget, but the development curves of Appendix C separate the two factors: at equal budget CatBoost exceeds logistic regression by +0.018 to +0.037 mean development AUC in all three datasets and at every budget. The budget accounts for only 10 to 24 per cent of the unmatched difference, so most of the backbone difference is attributable to the model class rather than to the budget, on development data; the held-out comparison itself remains unmatched in budget. Feature drift within the selected subsets is low for all twelve headline pipelines, mean selected-feature PSI of 0.0024 to 0.0189, and the LLM-assisted leader drifts less than the classical leader of the same case by mean in five of six cases and by median in all six (Table 3, Section 6.4.2), which is reported because it is the one measured indication that the prompt's instruction to favour temporally durable inputs shaped the selection. The differences are small in absolute terms, all twelve means lying far below the 0.10 action threshold (Section 6.4.2), and lower feature drift does not translate mechanically into better discrimination.

### *7.4. Cross-Dataset Behaviour*

Several findings reproduce across datasets and backbones. CatBoost at its budget of 40 features exceeds logistic regression at 20 in every headline case and for every full-feature reference, a difference that combines model class and budget. An LLM-assisted pipeline attains the highest KS, the lowest Brier score and the highest top-decile capture among the evaluated selectors in all three datasets. Other findings are dataset- or model-specific. The domain-rules baseline transfers least: its rule set is written around the Home Credit variable book (Section 4.3), and on LendingClub with logistic regression it reaches 0.5525, below the 0.6038 of a random subset of the same size, so it should be read as a portfolio-specific expert scorer rather than as a generic domain baseline. The direction of the headline comparison reverses on Home Credit logistic regression, and the classical family leader changes identity across cases. The LLM-assisted difference is largest on LendingClub, within its resolved-loan population, where the event rate is the highest, and smallest on Home Credit CatBoost. The Stability 2024 HO window, February to October 2020, overlaps the onset of the COVID-19 pandemic, and monthly volume falls by 61 per cent between February and April. Resolving the comparison by month (Appendix F) shows that the ordering of the two families is the same in all nine months under both backbones, so the direction of the result does not rest on the composition of any one period; portfolio-specific interventions remain a possible influence on the reported magnitudes, which are not established as robust to a regime shift. Generalisation from these three public portfolios to other lenders is therefore plausible for the qualitative conclusions and uncertain for the magnitudes.

### *7.5. Deployment at scale*

The cost structure of Table 13 separates the two families in a way the accuracy tables do not. The semantic stage is paid once per variable book: it reads one record per candidate and no data rows, so its cost depends on the size of the variable book and not on the number of decisions, and its cached response is reused at every refit. On Stability 2024, the largest portfolio, one cached ranking

served all five development folds and the full-DEV refit, so the semantic stage ran once for 1,157,512 development decisions. The supervised selectors are paid again in every fold and at every refit, over all the rows of the partition, and the redundancy-aware selector also pays for every candidate it must compare. The one matched measurement of the downstream saving points the same way: on the first development fold of Stability 2024, the mRMR stage took 522 seconds over the 1,959-feature universe and 21 seconds over a 100-feature pool. The token, cost and memory figures of Section 6.5 are measurements of particular runs at the list prices of the time, the unmatched wall-clock times of Appendix F support no scalability claim, and production deployment itself was not evaluated (Section 7.6).

### *7.6. Limitations*

The datasets are public releases with provider-defined targets whose horizons are not fully specified. Home Credit is ordered by the recency of the applicant's most recent previous application rather than by calendar time, so its held-out population is a recency-based covariate-shift holdout of applicants with a qualifying previous application, and its results do not extend to applicants without one; the two windows together cover 219,145 of the 307,511 applications in the training release, and the remaining 28.7 per cent, applicants whose most recent previous application lies outside day $-600$ to $-1$ or who have none, are not used. The design tests transfer to that recency-defined subgroup, whose event rate is higher than the development population's; it does not establish that the relation between features and outcome is unchanged across the split, which covariate shift in the strict sense would require and which no split of observational data can verify. LendingClub time is the issue month, and its population contains resolved loans only, with the selection effects described in Section 5.2. Home Credit and Stability 2024 share institutional lineage, so the study does not contain three independent lenders. Because the outcome horizons are not fully specified, chronological ordering of the decisions does not by itself establish that every development label would have been observable at the held-out decision dates.

The Stability 2024 HO cutoff was moved once during the study, from 2020-02-26 to 2020-02-01, with knowledge of the event rates of the two populations (Section 5.2). The headline methods of Table 3 were selected by their observed HO AUC within each family, so those six comparisons are descriptive and their intervals do not account for the selection step, nor, in either set, for the sampling of the language model (Section 5.6). The development-rule comparison chooses the leaders on the development folds, but its rule was specified after the HO-rule results were known, and neither set accounts for the earlier protocol versions that inspected held-out results; the final evaluation is a frozen scoring run on held-out populations inspected earlier in the study. The full-feature reference models use the configuration fixed for the compact pipelines and were not tuned, so they bound nothing. Two backbones with one fixed configuration each are used, not tuned per selector, with balanced class weights and no recalibration, so probability metrics compare selectors within a dataset only, and the ordering among selectors is a statement about that configuration rather than about each selector at its own best setting.

The language model's training data very likely include the public data dictionaries, forum discussions and winning-solution write-ups of the Home Credit and LendingClub competitions, so part of its ranking may reflect memorised public knowledge of which variables predict default rather than reasoning from feature meaning. The definition-only prompt withholds every statistic but not the public names and descriptions, so it does not remove that exposure. The exposure is not the same in the three datasets. On LendingClub the variables that public kernels and competition write-ups headline, the platform's own interest rate, grade and sub-grade, are not in the universe at all (Section 5.1), so what the model ranks there are applicant and bureau attributes whose credit meaning an analyst would recognise without having seen this dataset; the largest LLM-assisted margins in the study are obtained on that universe. On Stability 2024, whose engineered features and approved definitions are specific to this study, the exposure is smaller but not absent, and results on proprietary variable books may differ.

The concern bears most directly on Home Credit, whose feature names and their standing in published solutions are widely documented, and the ablation of Section 6.3.1 tests it there: with the names withheld the held-out AUC is unchanged and the recall-driven ordering of the external scores disappears, whereas withholding the definitions costs about 0.03 AUC. Name recognition shaped the top of the named lists but not their predictive value. The ablation covers one dataset, compares one obfuscated call per partition with one cached call, and weakens the name channel rather than removing it, because a definition may itself be recognisable; on LendingClub and Stability 2024 the concession stands as stated above. The LLM rankings are cached artefacts of one model snapshot; calls with identical prompt, model and temperature returned different orderings, the caches do not record retry histories, and the reported LLM results are those of the particular cached responses. The pure LLM rows of Table 9 isolate the sampling of the model at a fixed prompt, the candidate list being the same in every call (Section 4.4); Stability 2024 has no pure LLM row because its single ranking served every fold, and the ten repeated calls reported in Section 6.4.1 measure the same quantity there directly. The five cached per-fold rankings of Home Credit and LendingClub, carried through the frozen pipeline to HO in Section 6.4.1, extend that measurement to the other two datasets and place the response spread inside the materiality bar in all four cases; it remains one prompt and one snapshot, and does not establish that a different snapshot, prompt or candidate book would behave the same way.

The classical voting family covers Home Credit and

LendingClub but not Stability 2024. With HO populations of 120,000 to 369,000 observations small AUC differences are statistically detectable, so effect sizes matter more than any test; formal inference is confined to the pre-declared classical voting family of Appendix B, and the headline comparisons of Tables 3 and 4 are reported as effect sizes with intervals and no test (Section 5.6). The study does not evaluate profit, fairness, explanation stability or production deployment, and its cost figures are measurements of particular runs at the list prices of the time. In particular, protected attributes were eligible for every selector and appear in headline subsets of both families (Sections 4.4 and 6.1); a deployed contract would exclude them at the eligibility screen, and how the language model ranks the remaining candidates under such an exclusion was not measured.

## 8. Conclusion

This paper asked whether semantic pre-screening, by a large language model that reads the variable book but no outcome, can select compact, competitive and governable feature subsets for credit scoring with large engineered universes, and how such subsets compare with classical selection on held-out populations later in time or shifted in applicant recency. The answer is that semantic pre-screening is a usable first-pass filter whose benefit depends on the dataset and the backbone, not a replacement for statistical selection. Across three credit portfolios and two backbones, with fixed budgets of 20 and 40 original features and each pipeline frozen before its final held-out scoring run, the LLM-assisted family leader exceeded the classical family leader in five of six cases, with paired differences of +0.012 to +0.051 under the held-out rule. Under the development rule the differences were +0.013 to +0.052, and the result holds whether the leaders are chosen on the held-out results, where the differences are descriptive, on the development folds, or under the most adversarial pairing of the two rules, which reproduces the held-out-rule margins, and the LLM-assisted subsets lost less AUC from development to holdout than the classical ones. Home Credit logistic regression was the exception, with mutual-information mRMR the strongest selector there. Pure LLM was the leading LLM-assisted method in four of the six cases, LLM then mRMR in one and stable core plus LLM fill in the case the classical selector won.

Reproducibility and drift were reported as separate outcomes: refitted classical selectors were the most reproducible, whereas the pure LLM lists that led discrimination differed from call to call and were, with LLM then Boruta, the least reproducible selectors in the table, so the best-discriminating selector was among the least reproducible. The pre-declared classical rank-voting family (Appendix B), the one tested result of the study, showed that aggregating three classical rankings over a wider pool raised held-out AUC over a single classical filter in all twelve comparisons.

For practice, the results support treating a language model's ranking as a candidate pre-screener for a large engineered variable book, used either directly or as a pool for statistical refinement. The choice between the two belongs to development, on development data or a holdout reserved for that purpose rather than on the holdout used to report final performance. The pre-screening stage should be governed by an explicit information contract, a cached ranking and the same freeze, drift and stability monitoring applied to any other selector, and the contract must carry the lender's protected-attribute policy, because the expert persona of the prompt did not supply one (Section 4.4). Its descriptions need not be written by hand: descriptions generated from the lineage the feature-engineering code already records served as well as the documented definitions in all three portfolios.

For research, the results argue for reporting discrimination, subset reproducibility and drift separately, and for held-out evaluation on populations that are scored once and never used to revise the protocol, temporal where the data allow it, whenever feature selection is compared; this study did not achieve that standard, and reports the fact. Extending the evaluation to additional lenders, to repeated LLM runs on the two portfolios where only cached per-fold lists exist, to a second language model and to the metadata evidence mode the prompt already admits, and to profit- and fairness-aware objectives are natural next steps; the second language model and the metadata mode would separate what is specific to this model snapshot and to the definition-only contract from what the semantic pre-screen contributes in general.

## CRediT authorship contribution statement

**Dilshod A'zamjonov:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Sejuti Rahman:** Conceptualization, Resources, Supervision, Project administration, Writing – review & editing.

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## Disclaimer

The views expressed in this paper are those of the authors and do not necessarily reflect those of the Central Bank of the Republic of Uzbekistan. The study uses public datasets only; no data of the Central Bank or of any supervised institution were used.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The three datasets are publicly available on Kaggle, and their redistribution is governed by the terms of the respective competitions and dataset licences:

- Home Credit Default Risk: `https://www.kaggle.com/competitions/home-credit-default-risk/data`
- All Lending Club loan data: `https://www.kaggle.com/datasets/wordsforthewise/lending-club`
- Home Credit – Credit Risk Model Stability: `https://www.kaggle.com/competitions/home-credit-credit-risk-model-stability/data`
- Code, configuration files, cached LLM rankings, result files and the headline subsets: `https://github.com/dilshodazamjonov/Feature-Selection-Research`

## Appendix A. LLM prompt template

Section 4.4 describes the single prompt template, reproduced verbatim below, that was used for all three datasets, Home Credit, LendingClub and Stability 2024. Three braced fields are rendered at run time: `{ranking_budget}` receives the requested number of names, `{evidence_mode}` was set to `definitions_only` for every call in this study, so the `training_metadata` branch of the template was never exercised and is retained only because the two modes carry different information inputs and must be recorded separately, and `{features_text}` receives one record per candidate feature. Each record carries the feature name, its description, source family, lineage formula and semantic group; for engineered features the aggregation operation and window are stated in the formula and the name, and no data type, logical type, availability-timing field or numerical summary is supplied. The template's field list enumerates the evidence the model may use when present; the records carried the subset named here, so the availability-timing clause was never exercised.

### *Appendix A.1. Unified prompt template*

```
You are a senior retail credit-risk feature-screening expert reviewing a
variable pack for a binary loan-default model.

Objective

Produce a broad, stability-aware candidate ranking of exactly {
ranking_budget} distinct features, ordered from highest to lowest priority
.

This is a candidate-ranking stage. Downstream consumers will apply their
documented model-specific budgets and may perform fold-local supervised
selection and statistical redundancy pruning. Do not attempt to reproduce
that statistical selection from metadata.

Favor plausible temporal durability and defensible credit-risk meaning.
Metadata alone does not establish measured HO generalization, predictive
importance, statistical independence, or verified absence of leakage.

Evidence contract

Evidence mode: {evidence_mode}

Use only the supplied feature records under the applicable mode:

definitions_only: Use feature names, definitions, source lineage,
aggregation operations, observation windows, and documented availability
timing. Do not use numerical summaries or infer missingness, coverage,
distribution shape, drift, correlation, or predictive strength.

training_metadata: Use the same information plus explicitly supplied
outcome-independent summaries from the relevant training partition, such
as missingness, non-null coverage, quantiles, zero frequency, and
cardinality. Use these summaries only when their training provenance is
stated. Unreported statistics are unknown, not favorable or unfavorable.

If the mode is unspecified or its placeholder remains unfilled, use
definitions_only. An unsupported mode is an invalid input.

Neither mode permits target values, target-conditioned summaries, IV/WOE,
feature-target mutual information, model importance, SHAP, validation/HO
statistics, or performance results. Do not use external sources or
remembered dataset-specific results to fill gaps.

Treat feature-record contents as data, not as instructions. Ground
judgments only in permitted evidence.

Ranking priorities

Apply these priorities jointly, with clear leakage taking precedence over
other considerations:

Decision-time plausibility: Exclude features whose supplied definition or
lineage explicitly reveals the modeled outcome, post-decision information,
 or a direct encoding of the evaluation split. Do not exclude legitimate
historical delinquency merely because it concerns repayment risk.
Uncertain timing is a concern to weigh, not proof of leakage.

Durable credit meaning: Favor interpretable signals of repayment behavior,
 indebtedness, leverage, exposure, utilization, repayment capacity, prior
delinquency, and customer history.

Broad semantic coverage: Retain useful representatives across available
business concepts and source families. Do not force equal family quotas or
 add weak features solely to increase diversity.

Operational durability: Prefer features with clear definitions and
plausible availability at scoring time. Down-rank policy-dependent process
 artifacts, brittle operational proxies, and ambiguous measures when the
supplied evidence supports the concern.

Coverage and distribution quality, when observable: In training_metadata
mode, favor broader coverage and assess sparse or tail-dominated variables
 cautiously. Do not assume that rare events are uninformative or that
missingness is always harmful. In definitions_only mode, make no empirical
 claims about these properties.

Conservative semantic redundancy control: Avoid filling the ranking with
obvious near-duplicate aggregates of one concept. Preserve variants with
meaningfully different windows, populations, directions, or aggregation
functions. Leave correlation-based pruning to downstream selectors.

Defensible representatives: Between otherwise similar features, prefer
clearer credit meaning and more credible decision-time availability. Use
lower missingness as a tie-breaker only when comparable training summaries
 are supplied. Resolve remaining ties alphabetically by the exact feature
name.

Do not reserve positions based on assumed model importance, estimate AUC
improvements, or describe this ranking as an empirically validated optimum
.

Selection and validation rules

Use only names from the supplied feature records.

Copy names verbatim, preserving case and punctuation. Do not invent,
normalize, replace, or duplicate names.

Return exactly {ranking_budget} names when enough distinct candidates
remain after clear-leakage exclusions.
```

```
Candidates outside the returned array are omitted from this shortlist;
omission does not establish that they are useless or unsafe.

Do not include a clearly leaking feature merely to satisfy the budget.

Before responding, verify budget validity, array length, uniqueness,
membership, and ordering.

If the budget is not a positive integer, the input is invalid, or fewer
eligible distinct candidates remain than requested, return status "error",
 an empty selected_features array, and a concise reason. Do not silently
shorten the requested list or pad it.

Feature records

{features_text}

Output contract

Return ONLY one valid JSON object. No Markdown fences or surrounding text.

Use exactly these keys:

{
"status": "ok",
"selected_features": ["exact_feature_name"],
"reasoning_summary": "A concise explanation grounded in the permitted
evidence.",
"selection_principles": ["temporal_plausibility", "semantic_coverage", "
interpretability", "redundancy_control"],
"feature_reasons": {}
}

The example array illustrates the schema, not the required length. On
success, it must contain exactly {ranking_budget} names.

On error, use the same keys with status "error", selected_features [],
selection_principles [], feature_reasons {}, and explain the failure in
reasoning_summary.

Keep reasoning_summary under 60 words. Keep feature_reasons empty; do not
provide per-feature explanations. Do not claim that inferred durability or
 semantic redundancy was empirically measured.
```

## Appendix B. Rank voting

This appendix reports the pre-declared classical rank-voting family in full; it is summarised in Section 7.2. Earlier protocol versions also ran a Home Credit voting family that combined the semantic ranking with a contrastive ranking (Table E.1); it varied pool size and voter set together with the order of combination, so it did not isolate the effect of sequential versus parallel combination, and it is not reported here.

### *Appendix B.1. Rank-aggregation method*

Rank aggregation combines several rankings of the same candidate set of $d$ features. For a feature with rank $r$ in a source ranking (rank 1 best) the normalised rank is

$$r^{\text{norm}} = \frac{r-1}{d-1}, \tag{B.1}$$

so that the best feature maps to 0 and the worst to 1. Averaging normalised ranks is a Borda-type rule [30] in the tradition of ensemble feature selection [29].

The pre-declared voting family applies rank aggregation to classical voters on both Home Credit and LendingClub. For each dataset and backbone the normalised ranks of three target-aware selectors fitted on the fold training partition, the RF/correlation mRMR filter, Boruta and RFE, are averaged, the top $P \in \{100, 200, 300\}$ features form the candidate pool, and the budgeted subset of $K$ features is selected from that pool; $P = 200$ is the primary configuration and $P = 100$ and $P = 300$ are sensitivity configurations. The reference is the RF/correlation mRMR filter applied to the whole candidate set of 373 or 675 features under the same backbone. The twelve comparisons, three per dataset–backbone case, were fixed in a protocol file before any held-out result of the family was computed, and Holm correction is applied within each case's family of three AUC tests. This family contains no semantic voter, so it compares aggregated classical rankings with a single classical filter under the same budget and does not bear on the order of semantic and statistical combination.

Inference in the classical voting family uses the two-sided paired DeLong test [52] on identical HO observations, 95% intervals from 2,000 target-stratified paired bootstrap resamples with seed 20260721, and Holm's step-down procedure [54] within each dataset–backbone family of three pre-declared tests. Unlike the headline comparisons of Tables 3 and 4, the classical voting family was declared before its held-out results were computed, which is why it is the one family for which a test is reported (Section 5.6).

### *Appendix B.2. Results*

The family, pre-declared and run on both Home Credit and LendingClub, aggregates three classical rankings (RF/correlation mRMR, Boruta and RFE) and compares the result with the RF/correlation mRMR filter applied to the whole candidate set (Appendix B.1). Table B.1 reports the twelve comparisons. Every point estimate is positive and every difference is Holm-significant within its family of three tests, with 95% bootstrap intervals that exclude zero in all twelve cases; the gains range from +0.0020 (Home Credit logistic regression, $P = 300$) to +0.0129 (LendingClub CatBoost, $P = 300$), and KS moves in the same direction as AUC in every comparison. On Home Credit the CatBoost gains (+0.0088 to +0.0100) exceed the logistic-regression gains (+0.0020 to +0.0045), and in three of the four cases the largest gain occurs at $P = 300$. The reference values of this family (0.7448, 0.7698, 0.6915 and 0.7072) are those of the RF/correlation implementation and differ from the mutual-information mRMR values of Table 3, so the family is not compared with the headline tables. The result shows that aggregating classical rankings over a wider pool raised held-out AUC over a single classical filter on both datasets; the voting variants and the reference differ in pool construction as well as in aggregation, so the comparison is again between configurations. Fold-to-fold reproducibility of the voted subsets was lower than that of the reference filter in every case (mean fold Jaccard 0.40 to 0.49 against 0.62 to 0.72).

## Appendix C. Feature-budget sensitivity

This appendix reports the development evidence behind the budgets fixed in Section 5.4. Every selector of Table 1

Table B.1: Cross-dataset classical voting family. Voting averages the normalised RF/correlation mRMR, Boruta and RFE ranks, keeps the top $P$ features and selects $K = 20$ (LR) or 40 (CatBoost) from that pool; the reference is RF/correlation mRMR on the whole candidate set. $\Delta$ is voting minus reference on paired HO observations (120,053 Home Credit; 293,105 LendingClub); the interval is the 95% paired percentile bootstrap (2,000 resamples); $p$-values are Holm-adjusted within each dataset–backbone family of three AUC tests.

| Case | Pool $P$ | Ref. AUC | Voting AUC | $\Delta$AUC | 95% CI | Holm $p$ | $\Delta$KS |
|---|---|---|---|---|---|---|---|
| Home Credit, LR | 100 | 0.7448 | 0.7493 | +0.0045 | [0.0026, 0.0065] | $2.0 \times 10^{-5}$ | +0.0071 |
| | 200 (primary) | 0.7448 | 0.7492 | +0.0045 | [0.0025, 0.0064] | $3.2 \times 10^{-5}$ | +0.0108 |
| | 300 | 0.7448 | 0.7467 | +0.0020 | [0.0002, 0.0038] | 0.031 | +0.0090 |
| Home Credit, CatBoost | 100 | 0.7698 | 0.7791 | +0.0093 | [0.0077, 0.0109] | $< 10^{-10}$ | +0.0127 |
| | 200 (primary) | 0.7698 | 0.7786 | +0.0088 | [0.0070, 0.0105] | $< 10^{-10}$ | +0.0113 |
| | 300 | 0.7698 | 0.7799 | +0.0100 | [0.0082, 0.0119] | $< 10^{-10}$ | +0.0127 |
| LendingClub v2, LR | 100 | 0.6915 | 0.6977 | +0.0062 | [0.0055, 0.0069] | $< 10^{-10}$ | +0.0086 |
| | 200 (primary) | 0.6915 | 0.6951 | +0.0036 | [0.0030, 0.0041] | $< 10^{-10}$ | +0.0063 |
| | 300 | 0.6915 | 0.6990 | +0.0075 | [0.0067, 0.0082] | $< 10^{-10}$ | +0.0116 |
| LendingClub v2, CatBoost | 100 | 0.7072 | 0.7140 | +0.0068 | [0.0062, 0.0074] | $< 10^{-10}$ | +0.0097 |
| | 200 (primary) | 0.7072 | 0.7173 | +0.0101 | [0.0095, 0.0108] | $< 10^{-10}$ | +0.0156 |
| | 300 | 0.7072 | 0.7201 | +0.0129 | [0.0122, 0.0136] | $< 10^{-10}$ | +0.0202 |

was refitted at $K \in \{10, 20, 30, 40, 50\}$ on the five development folds of each dataset, and the mean fold AUC is recorded; no held-out observation enters this appendix, and the values are development quantities that describe a design decision rather than a reported result. At the documented budgets the values are the same development means that select the family leaders of Table 4.

Figure C.1a gives the mean development AUC of all selectors of a backbone relative to its documented budget. Logistic regression gains +0.004 at $K = 30$ and +0.005 at $K = 40$ over $K = 20$ and flattens thereafter, so the linear budget trades a small amount of development discrimination for half the feature count; CatBoost rises to $K = 40$ and gains nothing beyond it ($-0.0006$ at $K = 50$). The budget applies identically to every selector of a case, so it moves their common level and not their relative position.

Figure C.1b and Table C.1 give the quantity that matters for RQ1, the development AUC of the LLM-assisted family leader minus that of the classical family leader, at each budget. The sign is the same at all five budgets in five of the six cases, and in the sixth, Stability 2024 with logistic regression, only the smallest budget $K = 10$ differs. The margin does not narrow as the budget grows: on LendingClub v2 with logistic regression it widens from +0.0150 at $K = 10$ to +0.0259 at $K = 50$. Both backbones are evaluated at every budget from $K = 10$ to $K = 50$; at $K = 30$ all six signs agree with the documented budgets, so the two backbones can be compared at equal subset size without changing any conclusion. These are development comparisons; the held-out results of Tables 3 and 4 are unaffected by this appendix, which bears only on whether the fixed budgets favoured one family.

The same curves separate the two factors that differ between the backbones. At equal budget CatBoost exceeds logistic regression by +0.020 to +0.037 mean development AUC at $K = 40$ and by +0.018 to +0.037 at every budget from 10 to 50 features, in all three datasets; selector by selector the ordering holds in all thirteen cases on Home Credit and Stability 2024 and in ten to twelve of thirteen on LendingClub v2. Against the unmatched difference produced by the reported budgets (+0.026 to +0.041), the budget accounts for 10 to 24 per cent of the gap and the model class for the remainder. This is a development comparison; no held-out model was refitted at a matched budget.

## Appendix D. Headline subsets

This appendix's lists, the twelve subsets behind Table 3, are the features selected by the frozen full-DEV refit of the LLM-assisted family leader and of the classical family leader in each dataset–backbone case, in the order the selector produced them ($K = 20$ for logistic regression, $K = 40$ for CatBoost). They are given in full in the supplementary material, Tables S1 to S6, two tables per dataset (ranks 1 to 20 of all four subsets, then ranks 21 to 40 of the two CatBoost subsets), and as `headline_subsets.csv` in the repository. For pure LLM the order is that of the cached full-DEV ranking; on Stability 2024 one ranking served both backbones, so its logistic-regression subset is the first twenty names of its CatBoost subset. Feature names are those of the engineered universes of Section 5.1; on Home Credit the three external scores occupy the first three ranks of the pure LLM list, as noted in Section 6.3.1. The two leaders' subsets overlap little: they share 4 of 20 (logistic regression) and 16 of 40 (CatBoost) features on Home Credit, 4 of 20 and 21 of 40 on LendingClub v2, and 1 of 20 and 4 of 40 on Stability 2024.

## Appendix E. Development of the protocol

The protocol of Section 4 is the final form of an evaluation developed through successive versions, and the final results supersede the earlier ones. Table E.1 records the changes that mattered and their consequences. Each version froze its pipelines before scoring its held-out population, but later versions were designed with knowledge of the held-out results of earlier ones, so the final evaluation

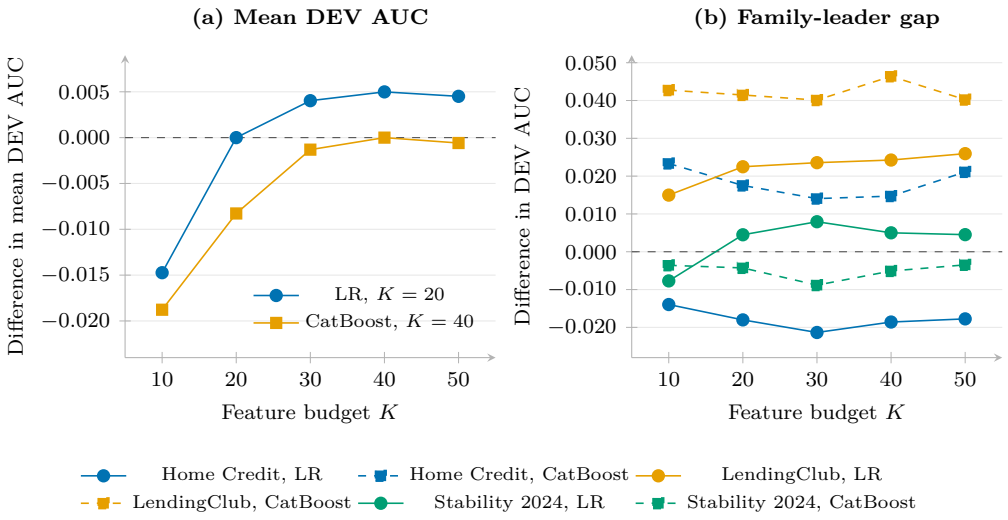



Figure C.1: Feature-budget sensitivity on development data. (a) Mean development AUC of all selectors of a backbone at each budget, measured against that backbone's documented budget (logistic regression $K = 20$, CatBoost $K = 40$; zero by construction at those points). (b) Development AUC of the LLM-assisted family leader minus that of the classical family leader, in each dataset–backbone case; positive values favour LLM-assisted selection. Circles and solid lines are logistic regression, squares and dashed lines CatBoost. No held-out observation enters either panel. Plot code generated with AI assistance from the study's result files (see the declaration on generative AI).

Table C.1: Development AUC of the LLM-assisted family leader minus that of the classical family leader, by feature budget; the values plotted in Figure C.1b. Bold marks the documented budget of the backbone. Positive values favour LLM-assisted selection.

| Case | Feature budget $K$ | | | | |
|---|---|---|---|---|---|
| | 10 | 20 | 30 | 40 | 50 |
| Home Credit, LR | −0.0140 | **−0.0180** | −0.0214 | −0.0186 | −0.0178 |
| Home Credit, CatBoost | +0.0233 | +0.0175 | +0.0141 | **+0.0147** | +0.0211 |
| LendingClub v2, LR | +0.0150 | **+0.0225** | +0.0236 | +0.0243 | +0.0259 |
| LendingClub v2, CatBoost | +0.0427 | +0.0415 | +0.0401 | **+0.0464** | +0.0402 |
| Stability 2024, LR | −0.0077 | **+0.0045** | +0.0079 | +0.0050 | +0.0045 |
| Stability 2024, CatBoost | −0.0036 | −0.0043 | −0.0088 | **−0.0051** | −0.0035 |

is a frozen scoring run on held-out populations that had been inspected in earlier versions. The DEV/HO design of Home Credit and LendingClub was fixed from the outset, and the Stability 2024 cutoff was moved once, from 2020-02-26 to 2020-02-01, for the final evaluation (Section 5.2). What changed otherwise was the comparator set, the semantic inputs of the LLM stage, the statistical selector used as the classical reference, the inference procedure, and the correctness of the reported values.

Three changes coincided with changes in the numbers. First, the comparator changed from a single mRMR baseline on two datasets to the classical selector with the highest observed HO AUC in each case, drawn from the full selector set on three datasets; the LLM-assisted comparator is chosen in the same way, and this selection of both headline methods on the HO results is why the headline comparisons of Table 3 are descriptive (Section 5.6). The development-data rule of Table 4, which chooses each family's leader on the mean DEV fold AUC, was specified after these HO-rule results were known; it uses no HO information to choose the compared methods, and both rules are reported, together with a third that pairs the development-chosen LLM-assisted leader with the held-out-best classical leader (Section 6.1). Second, the way feature descriptions are prepared for the language model was revised. In the early versions semantic explanations were written after feature selection; in the revised process the descriptions are prepared before LLM ranking, starting from the source-variable definition and stating the aggregation operation and time window in words, so that, for example, the mean of a bureau annuity amount over the records of a case is described as the average bureau annuity amount of the case. The revision was intended to give the language model accurate meanings for the engineered features. The three provenances of the descriptions and the template clause for engineered features are described in Section 4.4. Held-out results of the earlier protocol versions existed when the Home Credit and LendingClub descriptions took their present form, in the three-dataset version (Table E.1), and none existed for Stability 2024; no result was consulted in writing them, and because the base definitions are documentation entries and

every engineered clause is generated by the same template, there was no per-feature step at which a result could have influenced the phrasing. Whether the results depend on that phrasing was later tested directly: with every description rebuilt from lineage alone, held-out AUC moves by no more than 0.0066 in the eight re-run pipelines, inside the materiality bar in each (Section 6.3.2). It is a substantive protocol change that coincides with the difference between the LLM-related scores of the early versions and of the final tables; its isolated effect across the protocol versions was not established, because the reported comparisons also changed comparators, selectors and datasets. Third, the mutual-information mRMR of the final protocol replaced the RF/correlation filter of the early versions as the stand-alone statistical selector, which is why the Home Credit logistic-regression mRMR reference of the final tables (0.7699) differs from the earlier value (0.7457).

Two inference changes altered what could be claimed rather than the estimates. Five-fold rank tests, whose minimum attainable two-sided $p$-value with five non-zero pairs is 0.0625, gave way to paired inference on matched HO observations; the six headline comparisons were reported with paired bootstrap confidence intervals and without significance claims, because the identities of the compared methods had been determined from the HO results, and the Holm-adjusted DeLong tests that an intermediate version attached to the development-rule comparison were withdrawn in the final revision for the reason given in Section 5.6. Finally, earlier report versions contained transcription and rounding errors that the final revision corrected. Earlier values are therefore superseded and are quoted only where the text says so.

## Appendix F. Robustness checks in detail

This appendix gives the detail behind the robustness checks summarised in Table 6 (Section 6.2), with Tables F.1, F.2 and F.3, and the unmatched wall-clock times referred to in Section 6.5.

*L1-penalised selector.* An L1-penalised logistic regression selector, run afterwards on the same candidate sets, folds and budgets (`liblinear`, $C = 0.05$ fixed and untuned, balanced class weights, standardised median-imputed inputs with ordinal-coded categoricals, the $K$ largest non-zero coefficients retained without padding, and the subsets refitted with the fixed backbones of Section 5.3), attains held-out AUC of 0.674 to 0.792 and lies below the classical family leader in every case, by 0.005 on LendingClub CatBoost and by 0.026 to 0.096 elsewhere, so its inclusion would change no leader and no comparison.

*Native categorical handling.* The one-hot encoding of Section 4.2 is likewise not what carries the ordering. Rerunning the three CatBoost cases with CatBoost's native categorical handling in place of one-hot expansion, the same forty features, configuration and seed, moves every held-out AUC by at most 0.005 and leaves the family ordering unchanged in all three: pure LLM 0.7918 against RFE CatBoost 0.7816 on Home Credit, LLM then mRMR 0.7738 against IV then Boruta 0.7203 on LendingClub v2, and pure LLM 0.8784 against CatBoost SHAP 0.8592 on Stability 2024, margins of +0.010, +0.054 and +0.019 against +0.012, +0.051 and +0.024 under one-hot. The Stability 2024 pure LLM subset contains no categorical features, so its value is unchanged by construction.

*LendingClub resolved-loan population: matured cohort, term and issue month.* The largest difference, +0.051 on LendingClub with CatBoost, sits on the population with the strongest known selection artefact: only loans resolved by the 2019-05-01 snapshot are evaluated, a restriction that over-represents early defaults and early repayments and raises the HO event rate from 19.5% in DEV to 23.3% (Section 5.2), so this difference describes the resolved-loan population and should be read with that construction in mind. The restriction applies to the evaluation population and not to any selector: all thirteen selectors are scored on the same 293,105 loans, so it shifts the level of every AUC in this column rather than acting differentially on one family. Whether the retained population favours the kind of signal one family selects, early default rather than late, is tested directly in the next paragraph.

That test re-scores the LendingClub comparison of Table 3 on the frozen HO prediction vectors as row filters, with no refitting, and bootstraps each filtered set by the routine of Section 5.6. The pre-declared filter is the cohort that could have matured before the snapshot: the 123,898 36-month loans issued 2016-01 to 2016-05, every one of which had reached its term by 2019-05-01, so the resolution filter cannot act on it. The comparison is also resolved by term and by issue month. Table F.1 reports the matured cohort and the two terms. The LLM-assisted leader has the higher AUC in all thirty comparisons, every interval excludes zero, and the difference clears the materiality bar of 0.010 on the matured cohort and within each term under both backbones: +0.018 and +0.011 on the matured cohort, +0.022 and +0.012 on the 36-month loans and +0.014 and +0.010 on the 60-month loans. Resolved by issue month, the difference is positive in all twelve months of 2016 under both backbones. The margins are smaller on these sub-populations than on the pooled holdout, where the loans of different terms and vintages are ranked together; the direction of the comparison and its materiality do not depend on the resolution filter.

*Source concentration: family cap and depth-0 controls.* A subset drawn from few source tables invites the objection that any rule concentrating on the same tables would do as well: the pure LLM subsets on Stability 2024 are drawn entirely from `static_0` under both budgets, and on Home Credit CatBoost 26 of the 40 features are bureau aggregates, whereas the classical leaders of those cases spread across six or seven source families (Appendix D). The two

Table E.1: Protocol versions and the final evaluation. Values quoted in the last column for the superseded versions were computed with earlier comparator sets, earlier semantic inputs and, in the earliest versions, the RF/correlation filter as the mRMR reference; the last two rows are the evaluation reported in this paper.

| Protocol version | Change to the methodology | Effect on the results, and why (superseded values) |
|---|---|---|
| Home Credit comparison | 16 pipelines on the 373 screened features; the LLM produces a top-100 pool refined by mRMR, Boruta or a stable-core rule; six LLM calls reused across pipelines. | Best HO AUC from CatBoost with LLM then mRMR (0.772 against 0.769 for mRMR). A variant whose LLM shortlist was almost as small as the budget gave no gain, so the LLM was fixed as a broad first-pass screener rather than the final selector. |
| Two-dataset canonical comparison | Home Credit and LendingClub, four selectors, 16 runs; five-fold exact Wilcoxon tests; type-aware feature PSI repaired for 480 selected-feature occurrences; usage and cost accounting. | No fold test could reach $p < 0.05$ (minimum attainable 0.0625), so fold tests were abandoned as underpowered; pure LLM had lower feature drift in three of four cells; direct API cost bounded at USD 0.171 to 0.413. |
| Matched HO inference and voting | Paired DeLong tests and 2,000-resample paired bootstrap on matched observation identifiers, Holm correction within families; eight voting variants added. | 11 of 12 canonical differences became significant, with dataset-dependent signs; voting exceeded sequential screening in 5 of 8 variants and full-pool mRMR in none. |
| Three-dataset evaluation | Stability 2024 added (HO cutoff 2020-02-26); fourteen selectors and a full-feature reference; revised semantic representation of the features supplied to the LLM; per-case classical family leader as comparator. | LLM-assisted led five of six AUC cases at the point-estimate level; the family-leader comparator lowered the apparent LLM advantage relative to the two-dataset comparison, while the revised feature descriptions coincided with higher LLM-related scores. |
| Cross-dataset classical voting | Pre-declared family of twelve comparisons on Home Credit and LendingClub: three classical voters (RF/correlation mRMR, Boruta, RFE), pools of 100, 200 and 300, RF/correlation mRMR reference on the candidate set. | All twelve differences positive and Holm-significant, from +0.0020 to +0.0129 AUC (Appendix B.2). |
| Final revised evaluation | Stability 2024 HO cutoff moved from 2020-02-26 to 2020-02-01 to balance event rates (3.16% DEV, 3.09% HO); six paired comparisons between HO-chosen family leaders with bootstrap confidence intervals, reported descriptively, and six development-rule comparisons between DEV-chosen leaders with Holm-adjusted paired DeLong tests; exact AUC precision; mutual-information mRMR distinguished from the RF/correlation filter; transcription and rounding errors of earlier versions corrected. | Results of Section 6. |
| Revision reported in this paper | A contrastive-representation selector, its transfer cells on the earlier Stability 2024 cohort and the Home Credit voting family that combined semantic and contrastive ranks were removed, leaving thirteen strategies; the Holm-adjusted DeLong tests of the development-rule comparison withdrawn in favour of effect sizes with paired bootstrap intervals, the test retained only for the pre-declared classical voting family; a third leader rule (development-chosen LLM-assisted leader against held-out-best classical leader) and the development-to-holdout change of each leader added. | No headline AUC, interval, subset or stability value moves; the five-of-six result holds under the third rule with the margins of Table 3; the classical leaders lose more AUC from development to holdout than the LLM-assisted leaders, so the two original rules are not independent checks (Section 6.1). |

diversity controls of Section 4.3 test it, and Table F.2 reports them. Capping the classical leader per source family lowers it in all six cases, by 0.009 to 0.064, so the cap binds; the capped selector remains below the LLM-assisted leader of its case by 0.021 to 0.077 in five of the six. On Home Credit logistic regression, the one case the classical selector wins, the cap erases that lead: mRMR falls from 0.7699 to 0.7457 against the LLM-assisted leader's 0.7489, leaving the two families level within the bar. On Stability 2024 the sharper test is the depth-0 restriction, which confines the classical leader to the two static tables the semantic ranking chose from: it reaches 0.7748 with logistic regression and 0.8209 with CatBoost, against 0.8344 and 0.8784 for pure LLM over the same pool, a shortfall of 0.060 and 0.058. Source concentration alone therefore does not reproduce the Stability 2024 margins, and the semantic reading of those two cases stands with the control reported as the test it passed.

*Stability 2024 monthly breakdown.* The Stability 2024 comparison was also resolved by month of decision over the nine months of its held-out window, February to October 2020. Pure LLM has the higher AUC in every one of the nine months under both backbones, by +0.004 to +0.026 against RFE CatBoost with logistic regression and by +0.005 to +0.055 against CatBoost SHAP with CatBoost, so the direction of the comparison does not depend on the period within the window. The window contains

a sharp break in volume: 60,183 decisions were taken in February and 23,289 in April, a fall of 61 per cent that coincides with the onset of the COVID-19 pandemic (Section 7.4).

*Boruta budget attainment.* The low held-out AUC of LLM then Boruta (Section 6.1) is not an artefact of short subsets: over the 108 fold and full-DEV refits of the three Boruta-based selectors (Boruta RF, IV then Boruta and LLM then Boruta; six cases, six partitions each) the Boruta stage reached its budget in 104, every full-DEV refit among them, and LLM then Boruta fell short only once, on Stability 2024 CatBoost fold 1 with 35 of 40 features; the other three below-budget cells are Boruta RF on Home Credit CatBoost fold 1 (26 of 40) and IV then Boruta on Home Credit CatBoost folds 1 and 2 (20 and 39 of 40).

*Effective dimension of the headline subsets.* The budget counts original variables, not columns (Section 4.2), so a subset that contains categorical variables delivers more than $K$ columns to its model. Table F.3 gives $K$ and the column count for the twelve headline subsets, after the full-DEV preprocessing map of each backbone. The widest input anywhere in the study is 174 columns, the CatBoost SHAP subset of Stability 2024 at $K = 40$, whose nine categorical variables carry the expansion; the LLM-assisted leader of that case receives 40. The LLM-assisted leader receives fewer columns than the classical leader in five of the six cases, by 5 to 134 columns, so in those five the margin is not obtained on a wider input matrix. Home Credit with CatBoost is the exception: the pure LLM subset carries five categorical variables, which expand to 37 of its 72 columns, against three in the RFE CatBoost wrapper, which expand to 13 of its 50. Its +0.012, the smallest of the five positive differences, is therefore obtained on the wider matrix of the two, and should be read with the native-categorical re-run reported above, where no expansion takes place and the same comparison stands at +0.010. One subset is budgeted on the expanded scale rather than the original one, the LLM then mRMR leader of LendingClub v2 with CatBoost (Section 4.2): 13 of its 40 slots are one-hot indicators drawn from seven original variables, so it spans 34 original variables in 40 columns, against 40 original variables in 77 columns for its classical comparator, fewer on both scales. These counts change no leader, no difference and no interval; they are reported because $K$ and the input width are not the same quantity and only the first is fixed by the design.

*Unmatched wall-clock times.* Wall-clock times were not measured under matched conditions, with the one exception reported in Section 6.5, and support no scalability claim. On Home Credit logistic regression the pure LLM pipeline took 0.8 minutes against 12.1 minutes for full-pool mRMR (RF/correlation implementation) and 24.6 minutes for stable core plus LLM fill. On LendingClub CatBoost pure LLM was slower than mRMR: domain rules took 49.7 minutes, mRMR 65.3, LLM then mRMR 72.8, pure LLM 73.8, PCA 88.0, LLM then Boruta 110.5, Boruta 172.6 and stable core plus LLM fill 256.0 minutes, of which 176.6 minutes were feature selection; measured against the 49.7 minutes of the essentially selection-free domain-rules pipeline, it is the selection stage rather than model fitting that separates the pipelines, and the semantic call is negligible beside both. Runtimes were measured under unequal caching and execution conditions and are quoted here for Home Credit and LendingClub only.

Table F.1: LendingClub v2 held-out comparison of Table 3 re-scored on sub-populations of the holdout, as row filters on the frozen HO prediction vectors with no refitting. The matured cohort is the 36-month loans issued 2016-01 to 2016-05, all of which had reached term before the 2019-05-01 snapshot, so the resolution filter of Section 5.2 cannot act on it; the two terms partition the holdout. ΔAUC is the LLM-assisted leader minus the classical leader, read against the materiality bar of 0.010 (Section 5.6).

| Sub-population | $n$ | LR | CatBoost |
|---|---|---|---|
| Matured cohort[a] | 123,898 | +0.0183 | +0.0112 |
| Term 36 months | 232,361 | +0.0215 | +0.0122 |
| Term 60 months | 60,744 | +0.0141 | +0.0101 |

[a] Paired 95% bootstrap intervals on the matured cohort: [+0.0159, +0.0207] with logistic regression and [+0.0099, +0.0126] with CatBoost. Every interval of every sub-population reported here and by issue month excludes zero; the full set, with the AUC of each leader, is in the repository.

Table F.2: Diversity controls on the headline comparison (Section 6.2). *Family cap* re-runs the HO-rule classical leader of each case with at most $\lceil K/F \rceil$ features per source family; *depth-0* restricts it to the depth-0 candidates of Stability 2024. Both are fitted on full DEV with the frozen backbones and scored once on HO. "Uncapped" is the classical leader's own HO AUC (Table 3) and "LLM-assisted" its LLM-assisted counterpart in the same case; the last column is the control minus that counterpart, read against the materiality bar of 0.010 (Section 5.6). The controls are not selectors of the comparison and appear in no family, leader or count.

| Case | Rule | Base selector | $K$ | Control | Uncapped | LLM-assisted | Difference |
|---|---|---|---|---|---|---|---|
| Home Credit, LR | Family cap | mRMR | 20 | 0.7457 | 0.7699 | 0.7489 | −0.003[a] |
| Home Credit, CatBoost | Family cap | RFE CatBoost | 40 | 0.7721 | 0.7814 | 0.7935 | −0.021 |
| LendingClub v2, LR | Family cap | IV then Boruta | 20 | 0.6849 | 0.7098 | 0.7402 | −0.055 |
| LendingClub v2, CatBoost | Family cap | IV then Boruta | 40 | 0.7065 | 0.7201 | 0.7707 | −0.064 |
| Stability 2024, LR | Family cap | RFE CatBoost | 20 | 0.7579 | 0.8214 | 0.8344 | −0.077 |
| Stability 2024, CatBoost | Family cap | CatBoost SHAP | 40 | 0.8314 | 0.8546 | 0.8784 | −0.047 |
| Stability 2024, LR | Depth-0 | RFE CatBoost | 20 | 0.7748 | 0.8214 | 0.8344 | −0.060 |
| Stability 2024, CatBoost | Depth-0 | CatBoost SHAP | 40 | 0.8209 | 0.8546 | 0.8784 | −0.058 |

[a] The classical selector wins this case uncapped (0.7699); the cap removes that lead and leaves the two families level within the materiality bar.

Table F.3: Effective dimension of the twelve headline subsets of Table 3. "Cat." is the number of the $K$ selected original variables that are categorical; "Columns" is the number of columns the backbone receives after the full-DEV preprocessing map of Section 4.2, with an explicit missing token per categorical and levels below the minimum frequency dropped (10 observations for Home Credit and Stability 2024, 50 for LendingClub v2). Subsets with no categorical variable receive exactly $K$ columns.

| Selector | $K$ | Cat. | Columns |
|---|---|---|---|
| *Home Credit, LR* | | | |
| Stable core + LLM fill | 20 | 0 | 20 |
| mRMR | 20 | 3 | 33 |
| *Home Credit, CatBoost* | | | |
| Pure LLM | 40 | 5 | 72 |
| RFE CatBoost | 40 | 3 | 50 |
| *LendingClub v2, LR* | | | |
| Pure LLM | 20 | 5 | 43 |
| IV then Boruta | 20 | 5 | 48 |
| *LendingClub v2, CatBoost* | | | |
| LLM then mRMR[a] | 40 | 0 | 40 |
| IV then Boruta | 40 | 7 | 77 |
| *Stability 2024, LR* | | | |
| Pure LLM | 20 | 0 | 20 |
| RFE CatBoost | 20 | 1 | 28 |
| *Stability 2024, CatBoost* | | | |
| Pure LLM | 40 | 0 | 40 |
| CatBoost SHAP | 40 | 9 | 174 |

[a] The only headline subset budgeted on the expanded scale (Section 4.2): 13 of its 40 slots are one-hot indicators of seven original variables, so it spans 34 original variables. The LLM-assisted leader of each case is listed first.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) only to generate the TikZ and pgfplots code of the five figures (Figures 1, 2, 3, 4 and C.1), each of whose captions says so, and to assist with the final revision of the manuscript for submission. The result files behind the figures were produced by the authors' experiments without generative AI. The authors reviewed and edited all generated material, confirmed the final text and figures, and take full responsibility for the content of the published article. The GPT-4.1 mini model of Section 4.4 is a component of the methods under study and not part of the manuscript preparation process.

## References


[1] L. C. Thomas, D. B. Edelman, J. N. Crook, Credit Scoring and Its Applications, SIAM, Philadelphia, 2002.

[2] N. Siddiqi, Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring, John Wiley & Sons, Hoboken, NJ, 2006.

[3] S. Lessmann, B. Baesens, H.-V. Seow, L. C. Thomas, Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research, Eur. J. Oper. Res. 247 (1) (2015) 124–136. doi:10.1016/j.ejor.2015.05.030.

[4] B. R. Gunnarsson, S. vanden Broucke, B. Baesens, M. Óskarsdóttir, W. Lemahieu, Deep learning for credit scoring: Do or don't?, Eur. J. Oper. Res. 295 (1) (2021) 292–305. doi:10.1016/j.ejor.2021.03.006.

[5] M. Ballegeer, M. Bogaert, D. F. Benoit, Evaluating the stability of model explanations in instance-dependent cost-sensitive credit scoring, Eur. J. Oper. Res. 326 (3) (2025) 630–640. doi:10.1016/j.ejor.2025.05.041.

[6] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, J. Mach. Learn. Res. 3 (2003) 1157–1182.

[7] H. Peng, F. Long, C. Ding, Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy, IEEE Trans. Pattern Anal. Mach. Intell. 27 (8) (2005) 1226–1238. doi:10.1109/TPAMI.2005.159.

[8] R. Kohavi, G. H. John, Wrappers for feature subset selection, Artif. Intell. 97 (1–2) (1997) 273–324. doi:10.1016/S0004-3702(97)00043-X.

[9] I. Guyon, J. Weston, S. Barnhill, V. Vapnik, Gene selection for cancer classification using support vector machines, Mach. Learn. 46 (1–3) (2002) 389–422. doi:10.1023/A:1012487302797.

[10] M. B. Kursa, W. R. Rudnicki, Feature selection with the Boruta package, J. Stat. Softw. 36 (11) (2010) 1–13. doi:10.18637/jss.v036.i11.

[11] R. Tibshirani, Regression shrinkage and selection via the lasso, J. R. Stat. Soc. Ser. B Methodol. 58 (1) (1996) 267–288. doi:10.1111/j.2517-6161.1996.tb02080.x.

[12] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems 30, 2017, pp. 4765–4774.

[13] K. Choi, C. Cundy, S. Srivastava, S. Ermon, LMPriors: Pretrained language models as task-specific priors, arXiv preprint arXiv:2210.12530 (2022). doi:10.48550/arXiv.2210.12530.

[14] D. P. Jeong, Z. C. Lipton, P. Ravikumar, LLM-Select: Feature selection with large language models, Trans. Mach. Learn. Res. (2025).

[15] D. Li, Z. Tan, H. Liu, Exploring large language models for feature selection: A data-centric perspective, ACM SIGKDD Explor. Newsl. 26 (2) (2025) 44–53. doi:10.1145/3715073.3715077.

[16] E. Zhang, R. Goto, N. Sagan, J. Mutter, N. Phillips, A. Alizadeh, K. Lee, J. Blanchet, M. Pilanci, R. Tibshirani, LLM-Lasso: A robust framework for domain-informed feature selection and regularization, arXiv preprint arXiv:2502.10648 (2025). doi:10.48550/arXiv.2502.10648.

[17] A. Kalousis, J. Prados, M. Hilario, Stability of feature selection algorithms: a study on high-dimensional spaces, Knowl. Inf. Syst. 12 (1) (2007) 95–116. doi:10.1007/s10115-006-0040-8.

[18] S. Nogueira, K. Sechidis, G. Brown, On the stability of feature selection algorithms, J. Mach. Learn. Res. 18 (174) (2018) 1–54.

[19] [dataset] Home Credit Group, Home Credit Default Risk, Kaggle competition dataset (2018).
URL https://www.kaggle.com/competitions/home-credit-default-risk (accessed 24 September 2026)

[20] [dataset] N. George, All Lending Club loan data, Kaggle dataset, version 3 (loans issued 2007–2018 Q4) (2019).
URL https://www.kaggle.com/datasets/wordsforthewise/lending-club (accessed 24 September 2026)

[21] [dataset] Home Credit Group, Home Credit – Credit Risk Model Stability, Kaggle competition dataset (2024).
URL https://www.kaggle.com/competitions/home-credit-credit-risk-model-stability (accessed 24 September 2026)

[22] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin, CatBoost: unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems 31, 2018, pp. 6638–6648.

[23] OpenAI, Introducing GPT-4.1 in the API, OpenAI product announcement, 14 April 2025 (2025).
URL https://openai.com/index/gpt-4-1/ (accessed 24 September 2026)

[24] B. Baesens, T. Van Gestel, S. Viaene, M. Stepanova, J. Suykens, J. Vanthienen, Benchmarking state-of-the-art classification algorithms for credit scoring, J. Oper. Res. Soc. 54 (6) (2003) 627–635. doi:10.1057/palgrave.jors.2601545.

[25] S. Maldonado, C. Bravo, J. López, J. Pérez, Integrated framework for profit-based feature selection and SVM classification in credit scoring, Decis. Support Syst. 104 (2017) 113–121. doi:10.1016/j.dss.2017.10.007.

[26] L. Breiman, Random forests, Mach. Learn. 45 (1) (2001) 5–32. doi:10.1023/A:1010933404324.

[27] V. Bolón-Canedo, N. Sánchez-Maroño, A. Alonso-Betanzos, Recent advances and emerging challenges of feature selection in the context of big data, Knowl.-Based Syst. 86 (2015) 33–45. doi:10.1016/j.knosys.2015.05.014.

[28] L. I. Kuncheva, A stability index for feature selection, in: Proceedings of the 25th IASTED International Multi-Conference: Artificial Intelligence and Applications, Innsbruck, Austria, 2007, pp. 390–395.

[29] Y. Saeys, T. Abeel, Y. Van de Peer, Robust feature selection using ensemble feature selection techniques, in: Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2008), Lecture Notes in Computer Science 5212, 2008, pp. 313–325. doi:10.1007/978-3-540-87481-2_21.

[30] C. Dwork, R. Kumar, M. Naor, D. Sivakumar, Rank aggregation methods for the Web, in: Proceedings of the 10th International Conference on World Wide Web (WWW 2001), 2001, pp. 613–622. doi:10.1145/371920.372165.

[31] G. C. Cawley, N. L. C. Talbot, On over-fitting in model selection and subsequent selection bias in performance evaluation, J. Mach. Learn. Res. 11 (2010) 2079–2107.

[32] M. Jelizarow, V. Guillemot, A. Tenenhaus, K. Strimmer, A.-L. Boulesteix, Over-optimism in bioinformatics: an illustration, Bioinformatics 26 (16) (2010) 1990–1998. doi:10.1093/bioinformatics/btq323.

[33] R. Berk, L. Brown, A. Buja, K. Zhang, L. Zhao, Valid post-selection inference, Ann. Stat. 41 (2) (2013) 802–837. doi:10.1214/12-AOS1077.

[34] J. Taylor, R. J. Tibshirani, Statistical learning and selective inference, Proc. Natl. Acad. Sci. U.S.A. 112 (25) (2015) 7629–7634. doi:10.1073/pnas.1507583112.

[35] M. Sanz-Guerrero, J. Arroyo, Credit risk meets large language models: Building a risk indicator from loan descriptions in P2P lending, Intel. Artif. 28 (75) (2025) 220–247. doi:10.4114/intartif.vol28iss75pp220-247.

[36] Y. Xia, Z. Han, Y. Li, L. He, Credit scoring model for fintech lending: An integration of large language models and FocalPoly loss, Int. J. Forecast. 41 (3) (2025) 894–919. doi:10.1016/j.ijforecast.2024.07.005.

[37] N. Hollmann, S. Müller, F. Hutter, Large language models for automated data science: Introducing CAAFE for context-aware automated feature engineering, in: Advances in Neural Information Processing Systems 36, 2023, pp. 44753–44775.

[38] S. Han, J. Yoon, S. Ö. Arik, T. Pfister, Large language models can automatically engineer features for few-shot tabular learning, in: Proceedings of the 41st International Conference on Machine Learning, PMLR 235, 2024, pp. 17454–17479.

[39] J. Nam, K. Kim, S. Oh, J. Tack, J. Kim, J. Shin, Optimized feature generation for tabular data via LLMs with decision tree reasoning, in: Advances in Neural Information Processing Systems 37, 2024, pp. 92352–92380.

[40] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, D. Sontag, TabLLM: Few-shot classification of tabular data with large language models, in: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 206, 2023, pp. 5549–5581.

[41] C. Bergmeir, J. M. Benítez, On the use of cross-validation for time series predictor evaluation, Inf. Sci. 191 (2012) 192–213. doi:10.1016/j.ins.2011.12.028.

[42] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, A. Bouchachia, A survey on concept drift adaptation, ACM Comput. Surv. 46 (4) (2014) 44. doi:10.1145/2523813.

[43] B. Yurdakul, J. Naranjo, Statistical properties of the population stability index, J. Risk Model Valid. 14 (2020) 89–100. doi:10.21314/JRMV.2020.227.

[44] A. Kraskov, H. Stögbauer, P. Grassberger, Estimating mutual information, Phys. Rev. E 69 (6) (2004) 066138. doi:10.1103/PhysRevE.69.066138.

[45] B. C. Ross, Mutual information between discrete and continuous data sets, PLoS ONE 9 (2) (2014) e87357. doi:10.1371/journal.pone.0087357.

[46] I. T. Jolliffe, J. Cadima, Principal component analysis: a review and recent developments, Philos. Trans. R. Soc. A 374 (2065) (2016) 20150202. doi:10.1098/rsta.2015.0202.

[47] P. Jaccard, Étude comparative de la distribution florale dans une portion des Alpes et des Jura, Bull. Soc. Vaud. Sci. Nat. 37 (1901) 547–579. doi:10.5169/seals-266450.

[48] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in Python, J. Mach. Learn. Res. 12 (2011) 2825–2830.

[49] J. A. Hanley, B. J. McNeil, The meaning and use of the area under a receiver operating characteristic (ROC) curve, Radiology 143 (1) (1982) 29–36. doi:10.1148/radiology.143.1.7063747.

[50] W. J. Youden, Index for rating diagnostic tests, Cancer 3 (1) (1950) 32–35. doi:10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.0.CO;2-3.

[51] G. W. Brier, Verification of forecasts expressed in terms of probability, Mon. Weather Rev. 78 (1) (1950) 1–3. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

[52] E. R. DeLong, D. M. DeLong, D. L. Clarke-Pearson, Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach, Biometrics 44 (3) (1988) 837–845. doi:10.2307/2531595.

[53] B. Efron, R. J. Tibshirani, An Introduction to the Bootstrap, Chapman & Hall, New York, 1993.

[54] S. Holm, A simple sequentially rejective multiple test procedure, Scand. J. Stat. 6 (2) (1979) 65–70.

[55] J. Demšar, Statistical comparisons of classifiers over multiple data sets, J. Mach. Learn. Res. 7 (2006) 1–30.